

Sri Lanka Institute of Information Technology



Data Warehousing & Business Intelligence

Assignment 01

IT23346154

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1 Data Set Selection and OLTP Overview

1.1 Overview of the Selected Dataset

The dataset I selected for this assignment is a Retail Management and Customer Engagement System. This is an OLTP dataset that has six tables. These tables capture the day-to-day operations of a business. The dataset I have contains records from 2020 to 2024. This provides over four years of data. This exceeds the requirement of one year.

Dataset: [Retail Customer & Transaction Dataset](#)

All the data sources were initially provided in CSV format by the website. During the data preparation phase I converted the files into excel files. Imported them directly using a flat file source.

Final Format of Source Data Before Transformation:

1.1.1 CSV Files (.csv)

- customers.csv
- transactions.csv
- interactions.csv
- customer_reviews_complete.csv
- campaigns.csv

These CSV files were imported into SQL Server Management Studio (SSMS). Organized within a newly created database named Retail_SourceDB.

1.1.2 Excel File (.xlsx)

- support_tickets.xlsx

1.2 Solution Architecture

1.2.1 Retail_Staging

- StgCustomer
- StgCampaigns
- StgInteractions
- StgTransactions
- StgCustomerReviewsComplete
- StgSupportTickets

1.2.2 Retail_DW

- DimDate
- DimCustomer

- DimCampaign
- DimProduct
- FactSales
- FactInteractions
- FactSupport
- FactReview

1.3 Architecture Components

1.3.1 Data Source

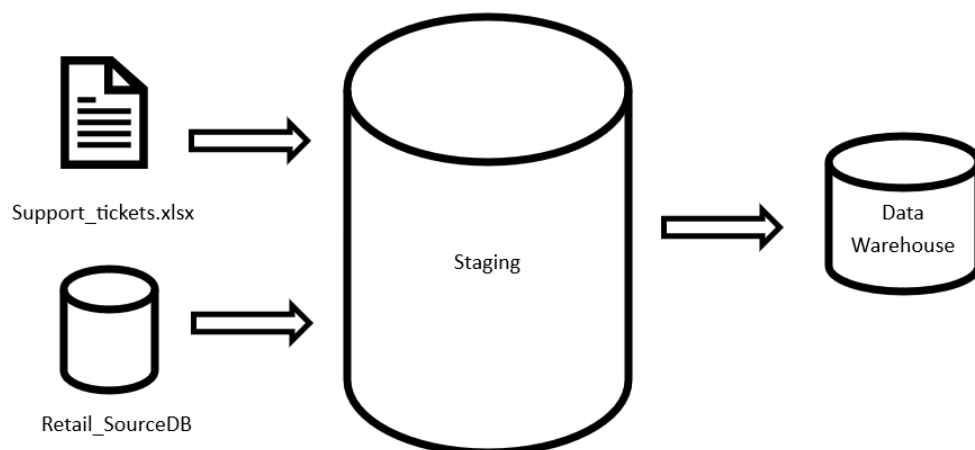
- Transactional processing systems (databases)
- Third -party data providers and external datasets

1.3.2 Extract, Transform and Load

- Extraction phase. Retrieving information from source database
- Conversion phase. Merging datasets eliminating duplicates
- Quality control. Standardizing and cleansing information

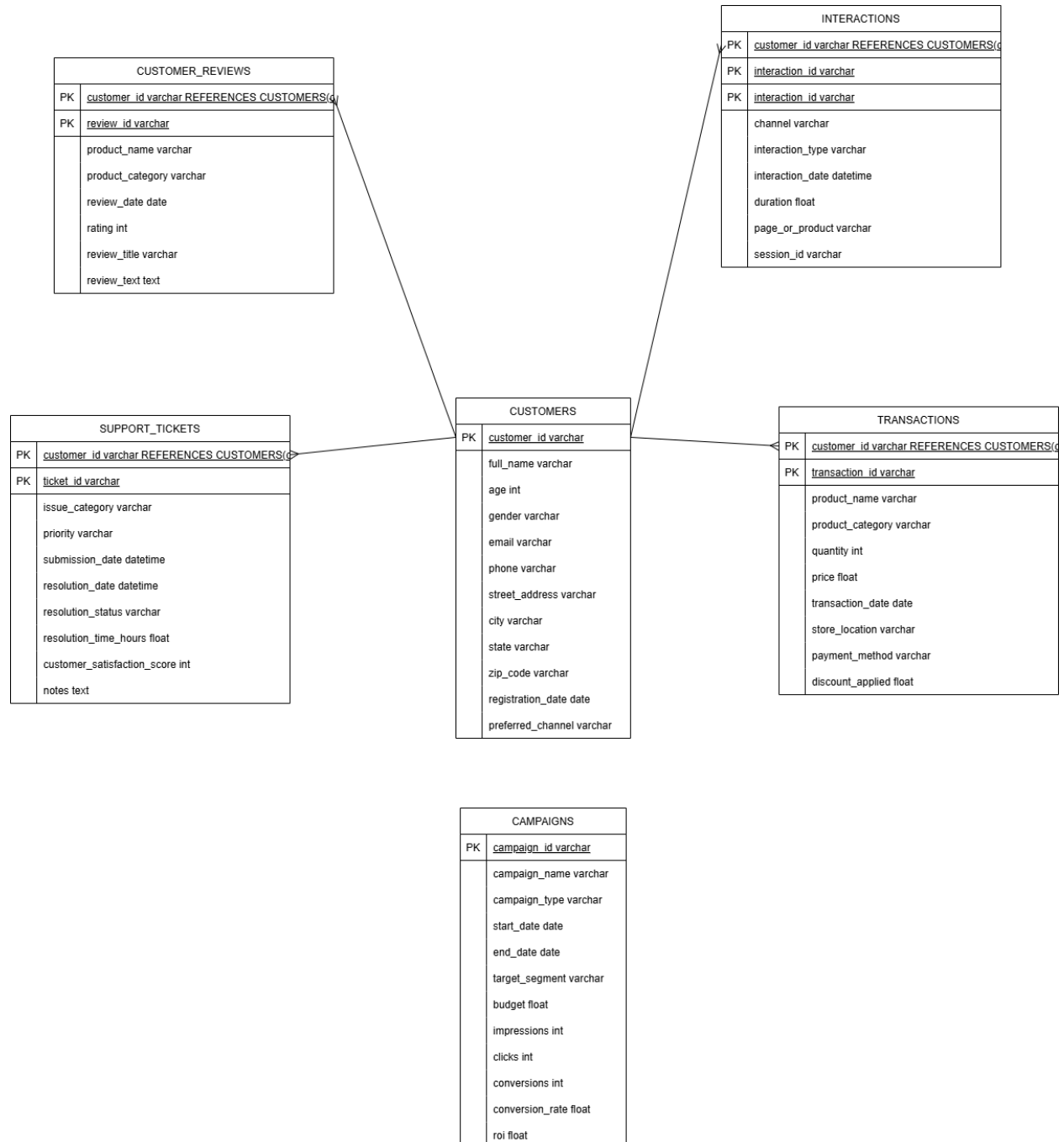
1.3.3 Data Warehouse

- EDW and Data Mart.
- Dimensional Modeling. Facts. Dimensions.
- Schema approach. Utilizing star-based structure for relationships



1.4 OLTP Entity Relationship Diagram (ERD)

The normalized structure of the source system is shown below. The Customers table is the entity. It is linked to activity tables via the customer_id primary key.



2 Dimensional Modeling

2.1 Star Schema Design

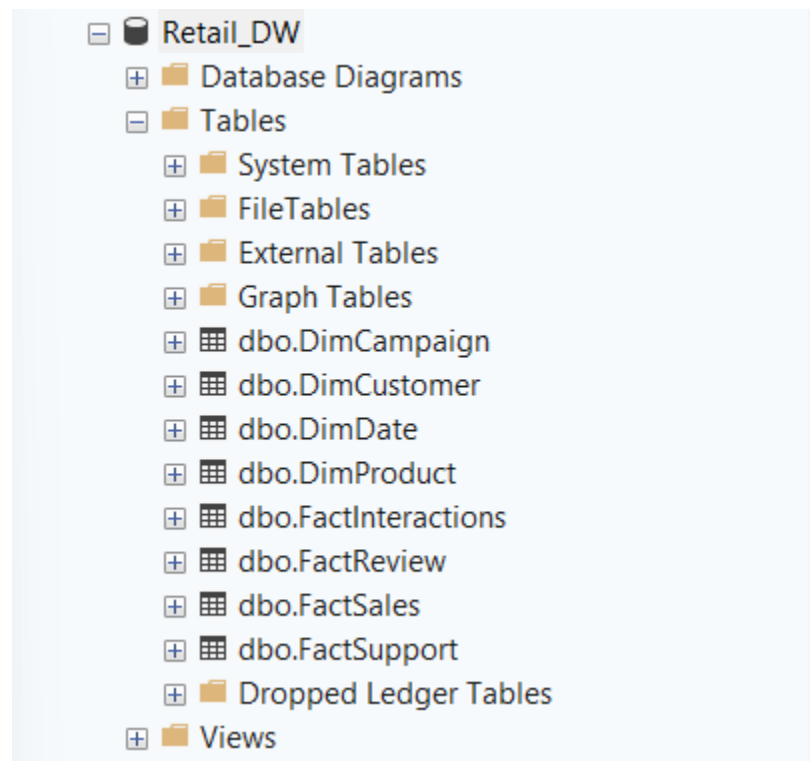
The OLTP data was de-normalized into a Star Schema to optimize it for processing (OLAP). This structure simplifies queries. It also improves performance for Business Intelligence reporting.

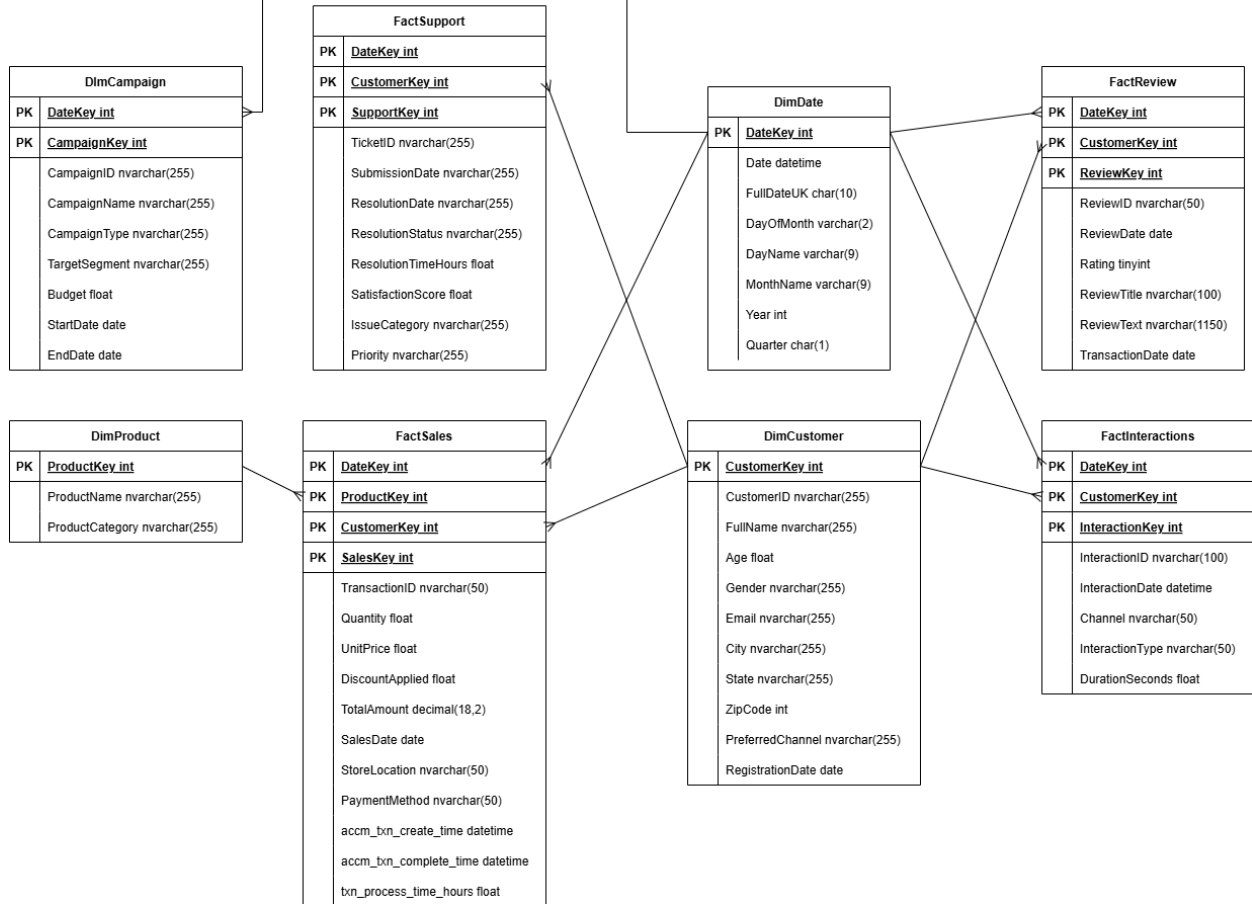
2.1.1 Dimension Tables:

- DimCustomer: Contains attributes like Name, Age, Gender, City, and State.
- DimProduct: Contains product hierarchies such as Product Name and Category.
- DimDate: Attributes for temporal analysis (Year, Quarter, Month, Day).
- DimCampaign

2.1.2 Fact Table:

- FactSales
- FactSupport
- FactInteractions
- FactReview





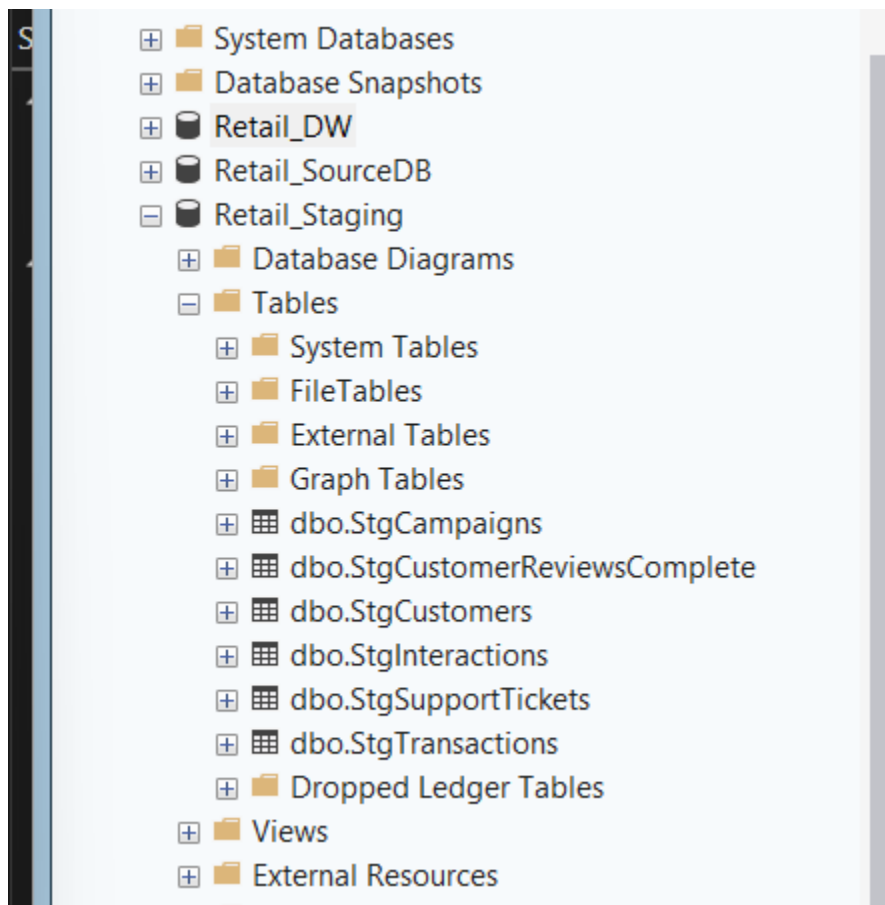
3 ETL Process Implementation (SSIS)

3.1 Data Extraction

Data was extracted from the provided Flat Files (xlsx format) using the Flat File Source component. I also extracted data from Source Data Base using OLE DB Source in SQL Server Integration Services (SSIS). Each source file and data was mapped to its staging table.

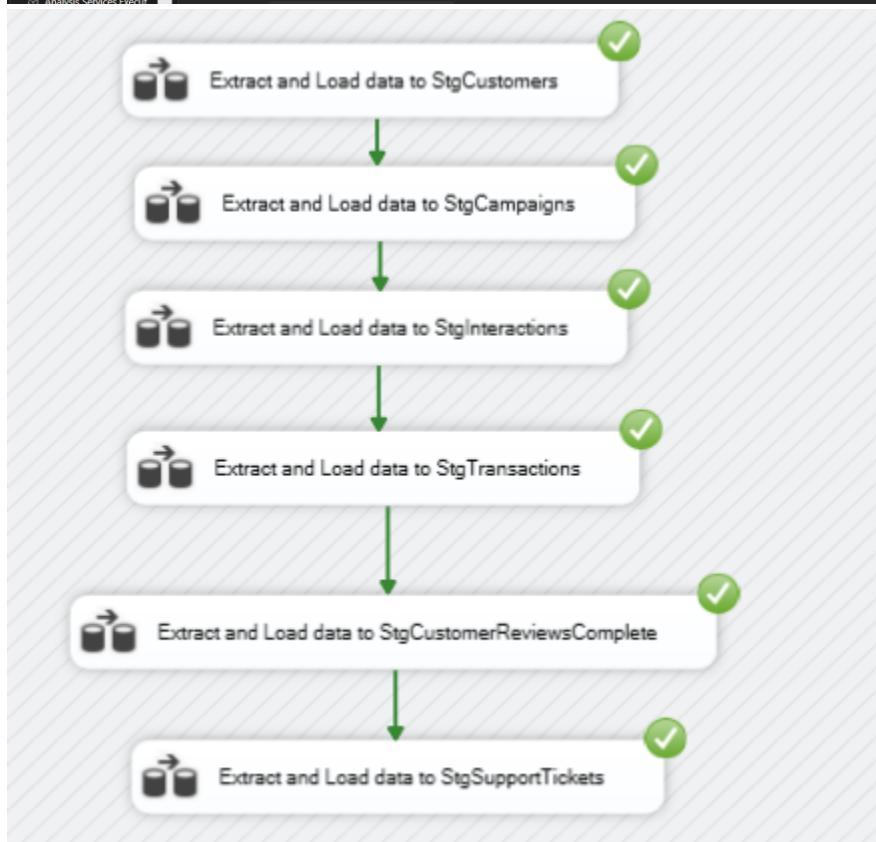
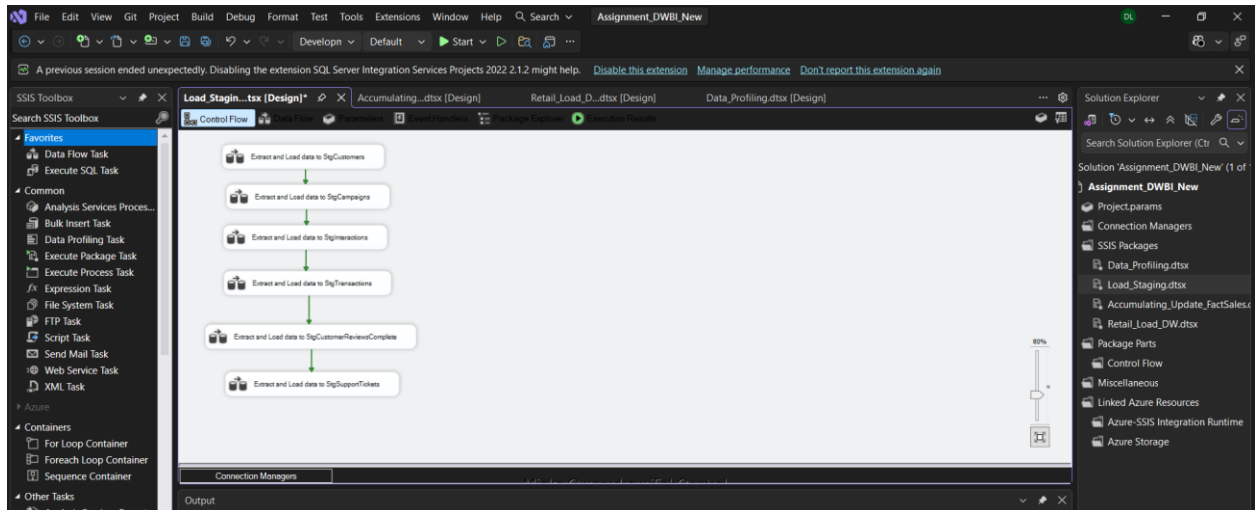
Staging tables :

- StgCampaigns,
- StgCustomerReviewsComplete,
- StgCustomer,
- StgInteractions,
- StgSupportTickets,
- StgTransactions



3.1.1 Load Staging Tables

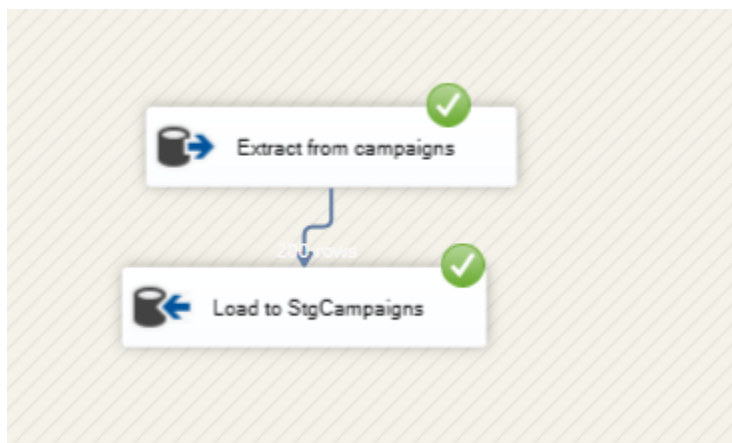
I created a SSIS Package Load_Staging.dtsx. This package extracts data from sources. Loads it to the staging tables.



3.1.1.1 Load customer data to Staging



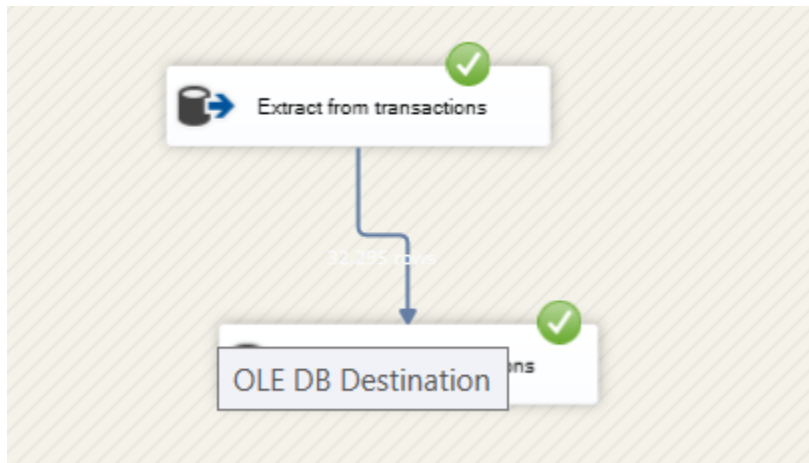
3.1.1.2 Load campaigns data to Staging



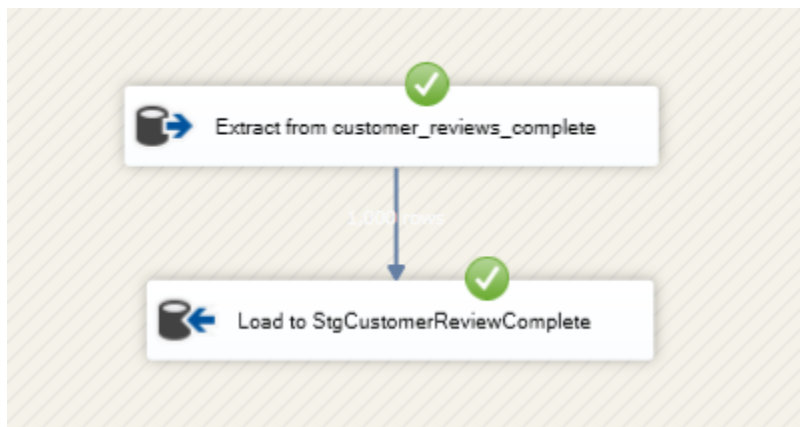
3.1.1.3 Load Interactions data to Staging



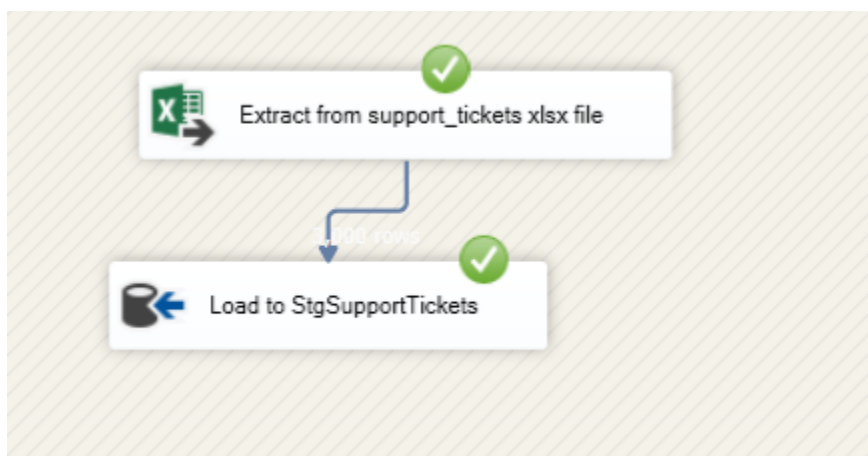
3.1.1.4 Load transactions data to Staging



3.1.1.5 Load customer reviews data to Staging

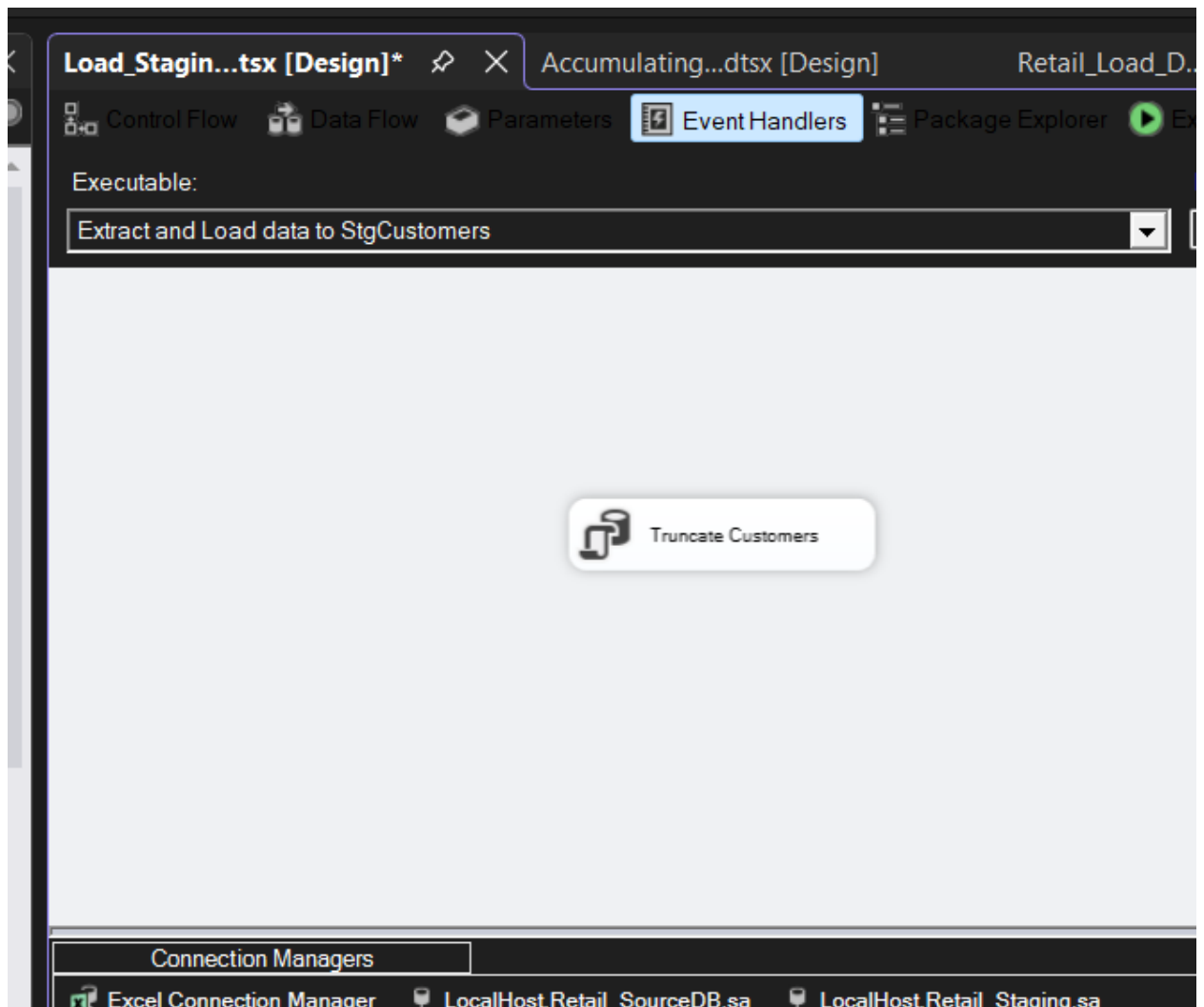


3.1.1.6 Load support tickets data to Staging

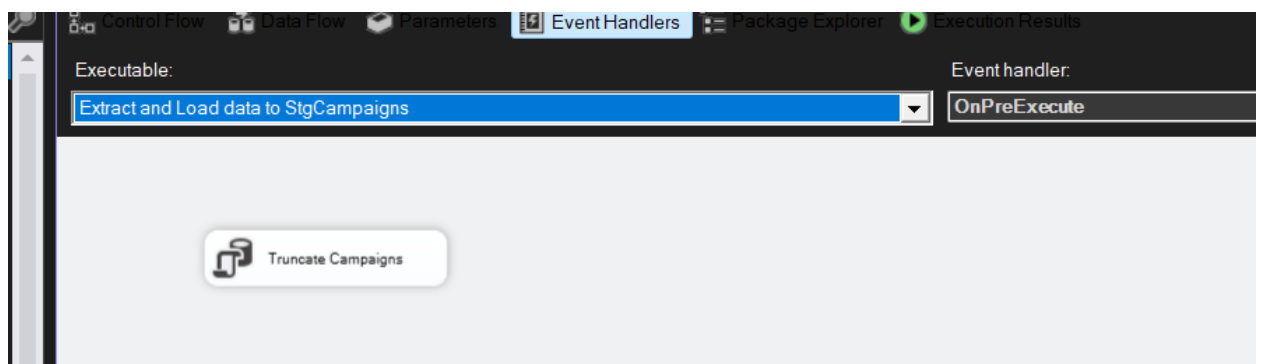


3.1.2 Truncated all Staging Tables to avoid duplicating values.

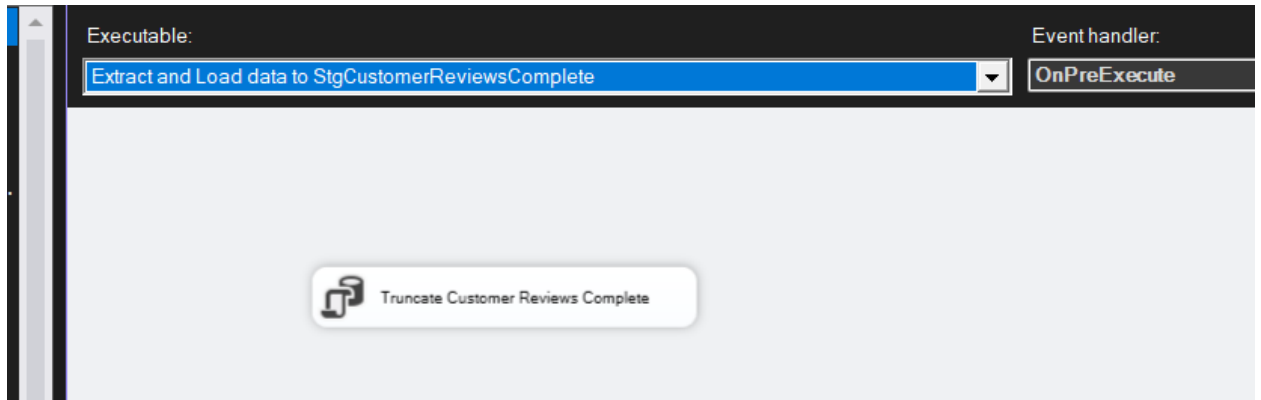
3.1.2.1 Truncate StgCustomers table



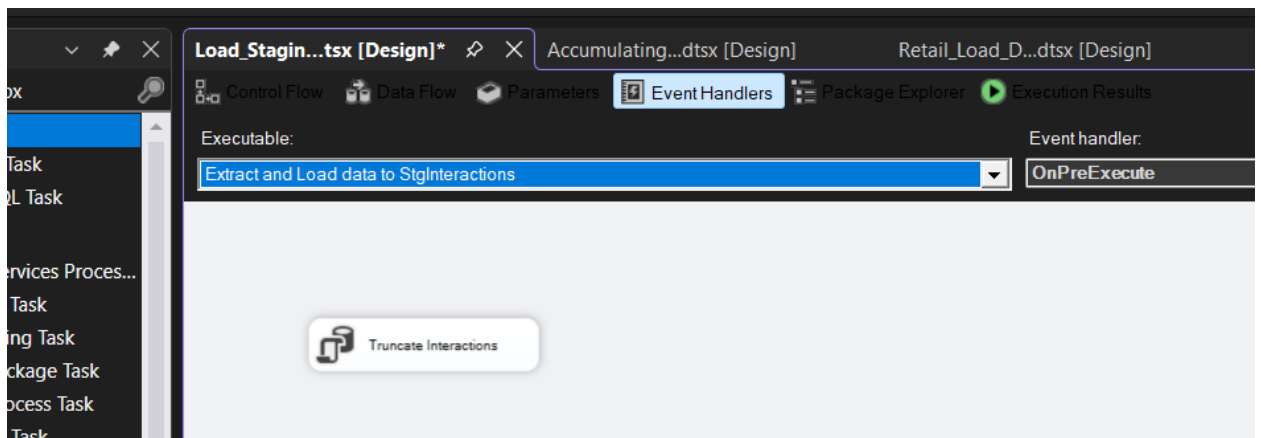
3.1.2.2 *Truncate StgCampaigns table*



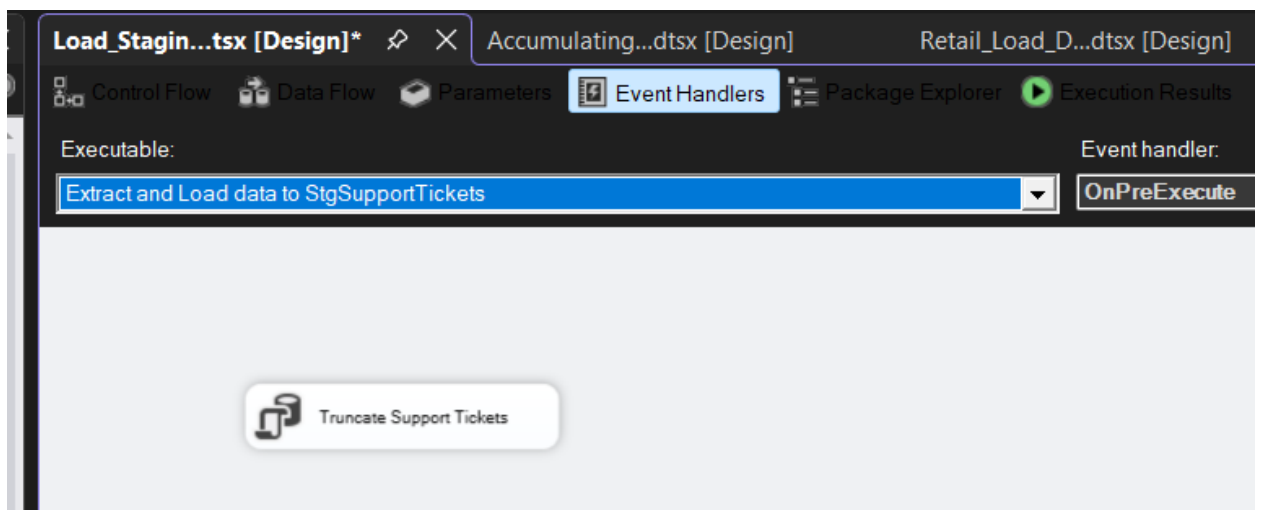
3.1.2.3 *Truncate StgCustomerReviewComplete table*



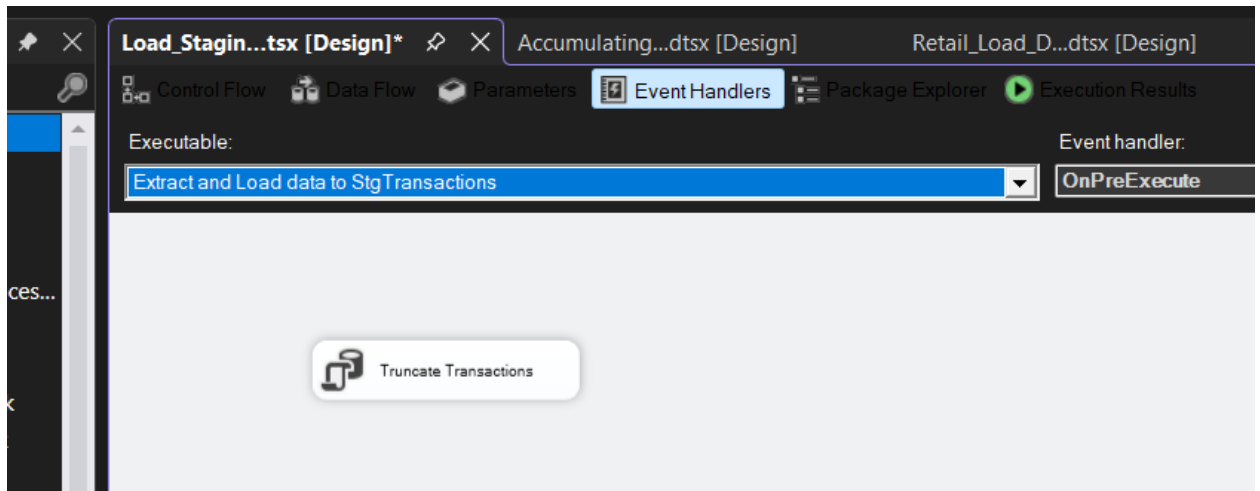
3.1.2.4 Truncate StgInteractions table



3.1.2.5 Truncate StgSupportTickets table



3.1.2.6 Truncate StgTransactions table



3.1.3 Data Profiling

3.1.3.1 Create data profiles for all staging tables



Name	
	Assignment_DWBI_New
	Campaigns_Profile
	Customer_Reviews_Complete_Profile
	Customers_Profile
	Interactions_Profile
	Support_Tickets_Profile
	Transactions_Profile

3.2 Data Transformation (T)

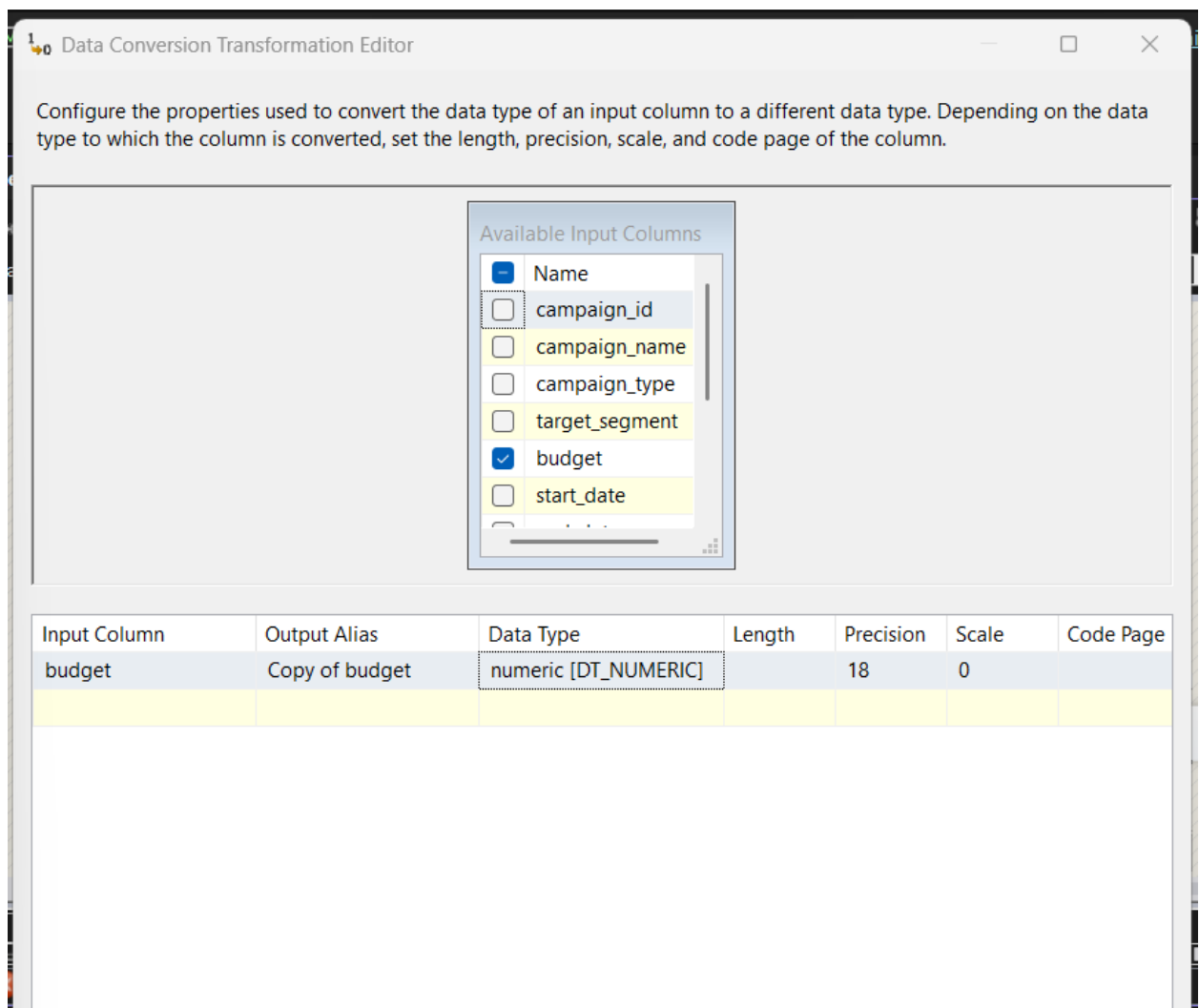
To clean and prepare the data for the warehouse, several transformations were applied:

Data Conversion

Converted CSV string data into appropriate SQL Server data types (e.g., Integer, Decimal, and Date).

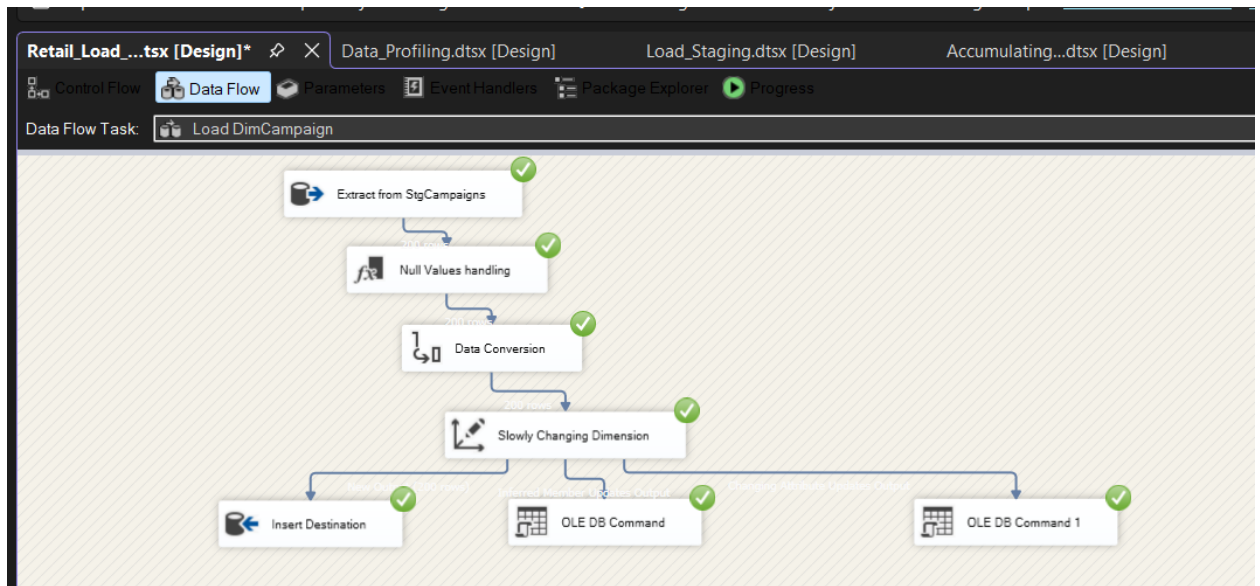
Budget column data type conversion in Load DimCampaign to numeric [DT_NUMERIC]

3.2.1.1 Data Conversion in Load DimCampaign Data Conversion Transformation



The screenshot shows the 'Data Conversion Transformation Editor' window. It contains a list of 'Available Input Columns' with checkboxes next to each column name. The 'budget' column is selected with a checked checkbox. Below the list is a table with columns: Input Column, Output Alias, Data Type, Length, Precision, Scale, and Code Page. The 'budget' row is highlighted, showing 'Copy of budget' as the output alias and 'numeric [DT_NUMERIC]' as the data type. The precision is set to 18 and the scale is set to 0.

Input Column	Output Alias	Data Type	Length	Precision	Scale	Code Page
budget	Copy of budget	numeric [DT_NUMERIC]		18	0	



3.2.2 Derived Column

A calculated column **TotalAmount** was created using the formula: (quantity * price - discount_applied) in **FactSales** table.

3.2.2.1 Calculate TotalAmount in Load to FactSales using Derived Column Transformation

Derived Column Transformation Editor

Specify the expressions used to create new column values, and indicate whether the values update existing columns or populate new columns.

Variables and Parameters

Columns

Mathematical Functions

String Functions

Date/Time Functions

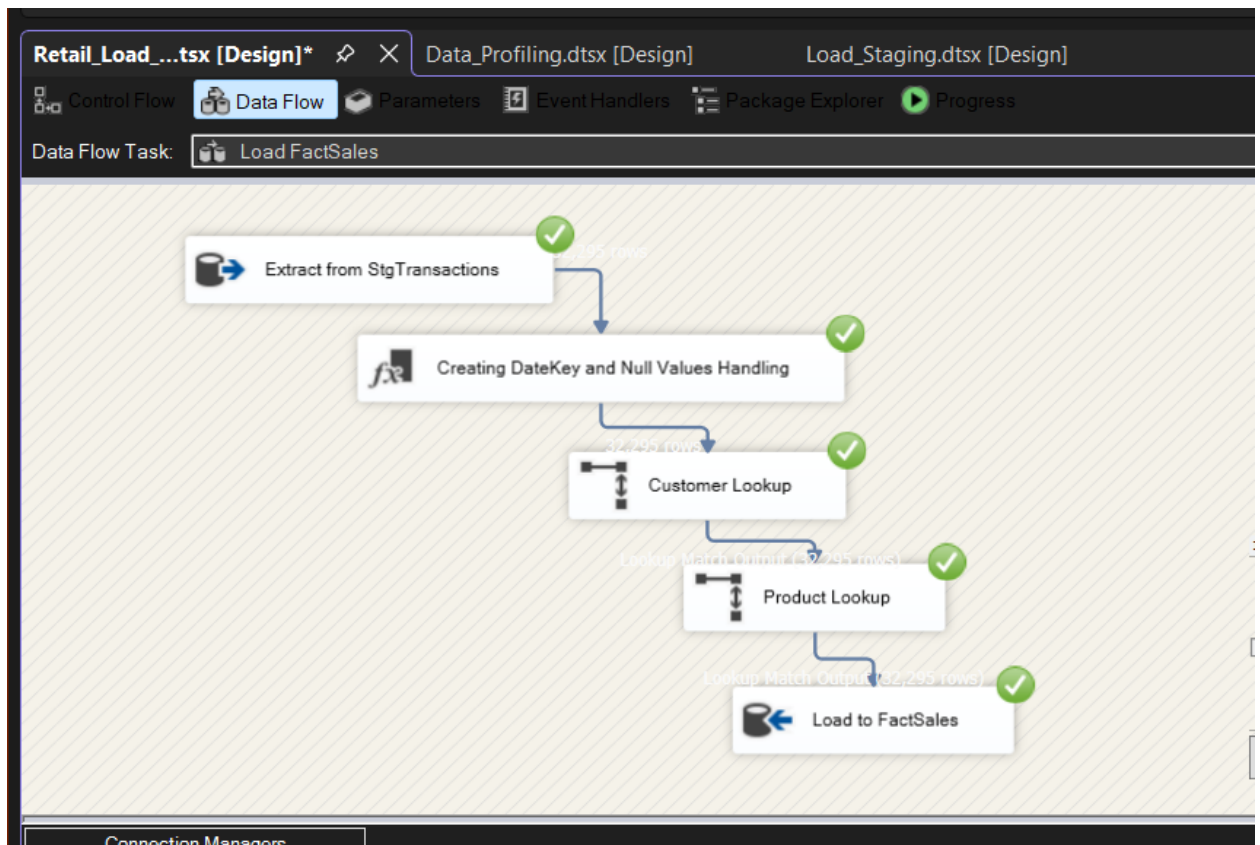
NULL Functions

Type Casts

Operators

Description:

Derived Column Name	Derived Column	Expression	Data Type	Le
DateKey	<add as new column>	(DT_I4)(REPLACE(SUBSTRING((DT_WSTR,50)tra...	four-byte signed inte...	
TotalAmount	<add as new column>	(quantity * price - discount_applied)	double-precision flo...	
quantity	<add as new column>	ISNULL(quantity) ? 1 : quantity	double-precision flo...	
price	<add as new column>	ISNULL(price) ? 0 : price	double-precision flo...	
discount_applied	<add as new column>	ISNULL(discount_applied) ? 0 : discount_appli...	double-precision flo...	
product_name	<add as new column>	ISNULL(product_name) ? "N/A" : product_name	Unicode string [DT_...	5
payment_method	<add as new column>	ISNULL(payment_method) ? "N/A" : payment_...	Unicode string [DT_...	5
store_location	<add as new column>	ISNULL(store_location) ? "N/A" : store_location	Unicode string [DT_...	5



Null values handing:

ISNULL(channel) ? "Unknown" : channel

ISNULL(interaction_type) ? "Unknown" : interaction_type

ISNULL(duration) ? 0 : duration

3.2.2.2 Handle NULL values in Load to FactInteraction using Derived Column Transformation

Derived Column Transformation Editor

Specify the expressions used to create new column values, and indicate whether the values update existing columns or populate new columns.

Variables and Parameters

Columns

interaction_date

channel

interaction_type

duration

interaction_id

customer_id

page_or_product

session_id

Mathematical Functions

String Functions

Date/Time Functions

NULL Functions

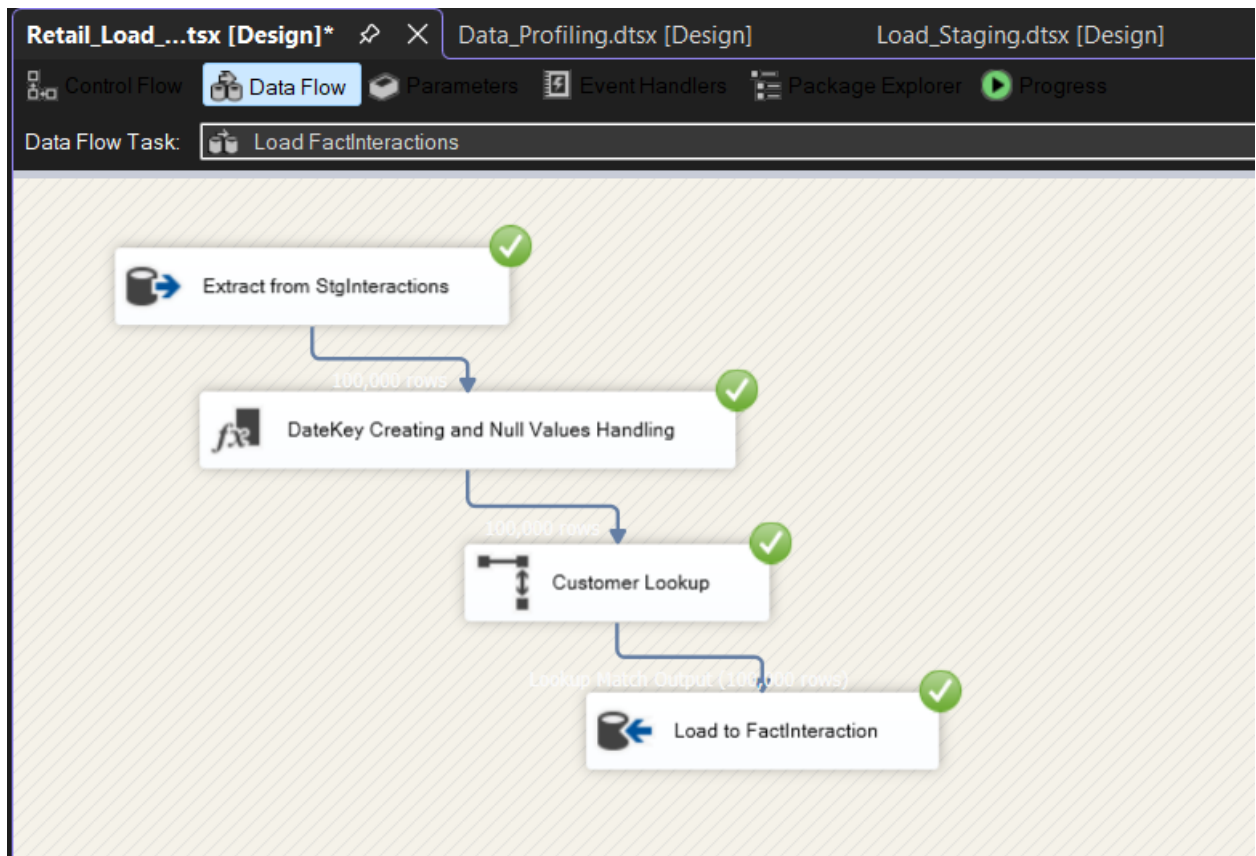
Type Casts

Operators

Description:

Derived Column Name	Derived Column	Expression	Data Type	Length
DateKey	<add as new column>	(DT_I4)(REPLACE(SUBSTRING((DT_WSTR,50)int...	four-byte signed inte...	
channel	<add as new column>	ISNULL(channel) ? "Unknown" : channel	Unicode string [DT_...	5
interaction_type	<add as new column>	ISNULL(interaction_type) ? "Unknown" : interac...	Unicode string [DT_...	5
duration	<add as new column>	ISNULL(duration) ? 0 : duration	double-precision flo...	

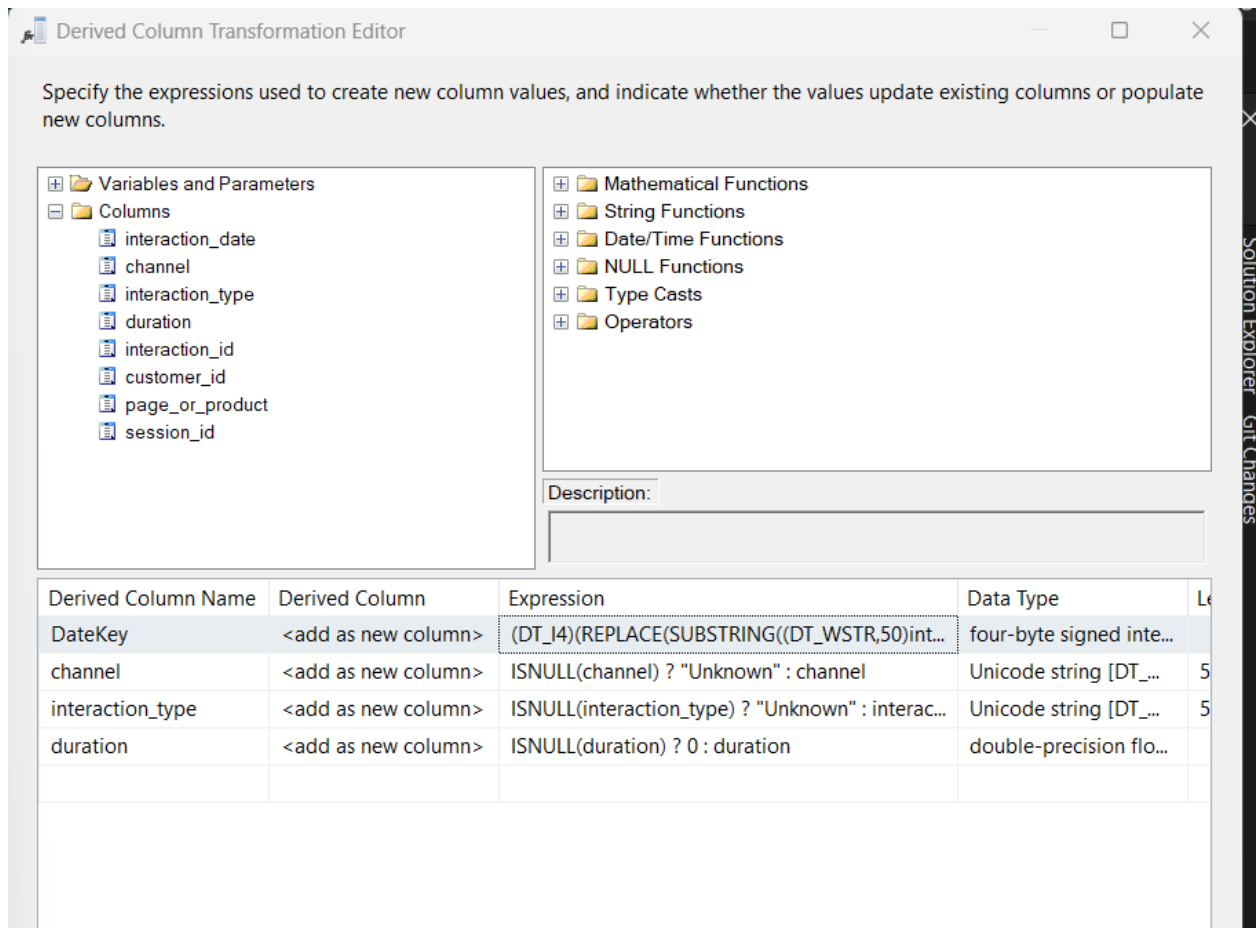
Solution Explorer - Git Changes



Creating DateKey:

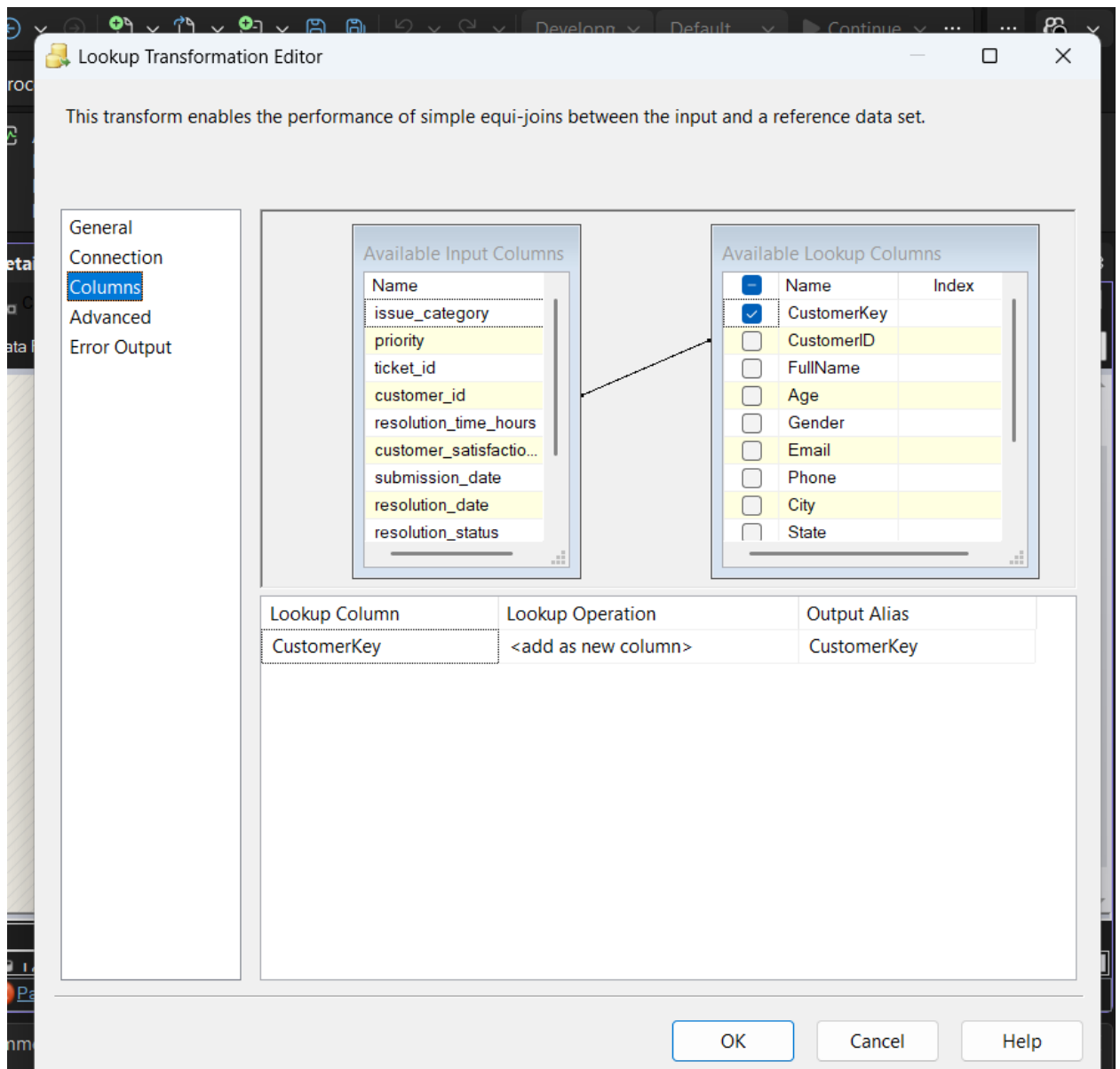
`(DT_I4)(REPLACE(SUBSTRING((DT_WSTR,50)interaction_date,1,10),'-', ''))`

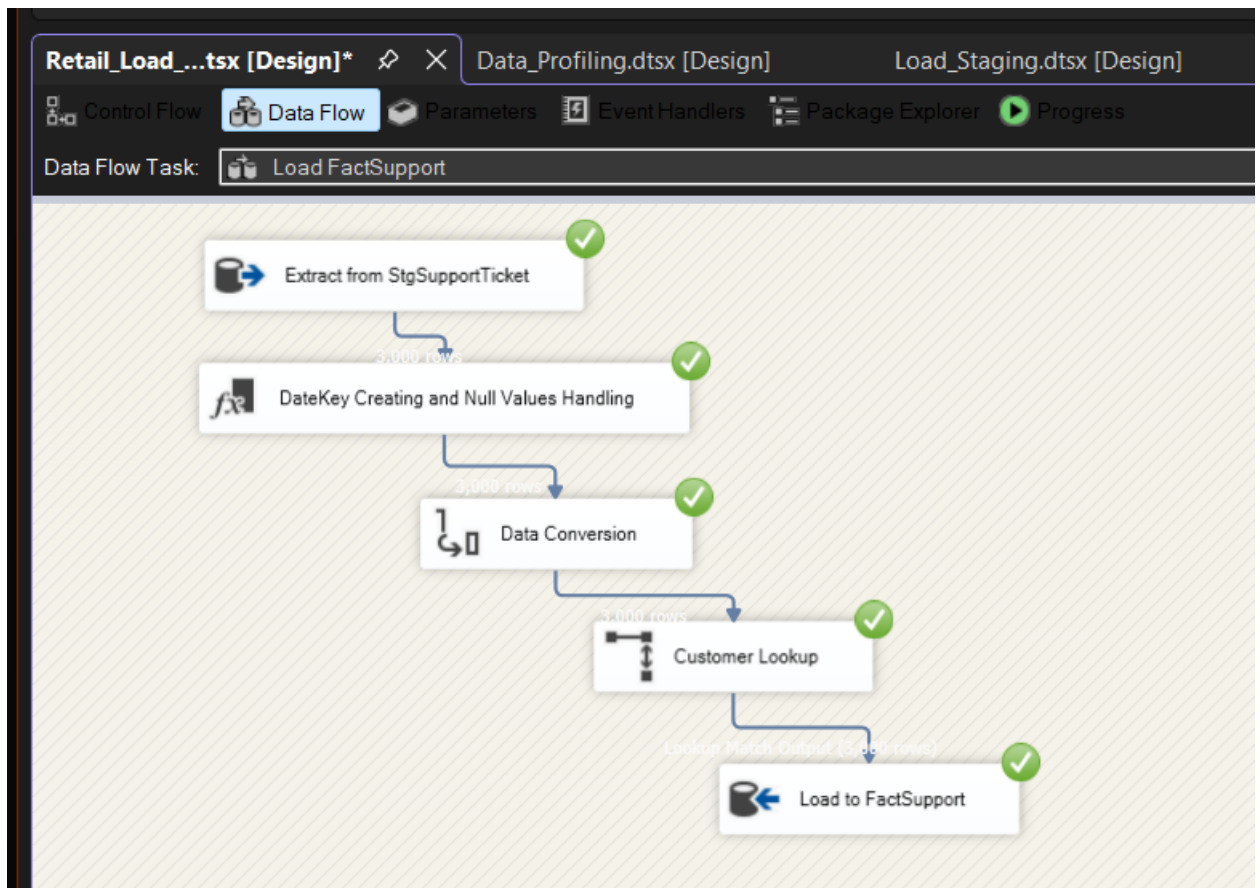
3.2.2.3 Create DateKey in Load to FactInteraction using Derived Column Transformation



3.2.3 Lookup Transformation

I mapped keys (like customer_id) to the surrogate keys in the Dimension tables. This maintains integrity in the Fact table.

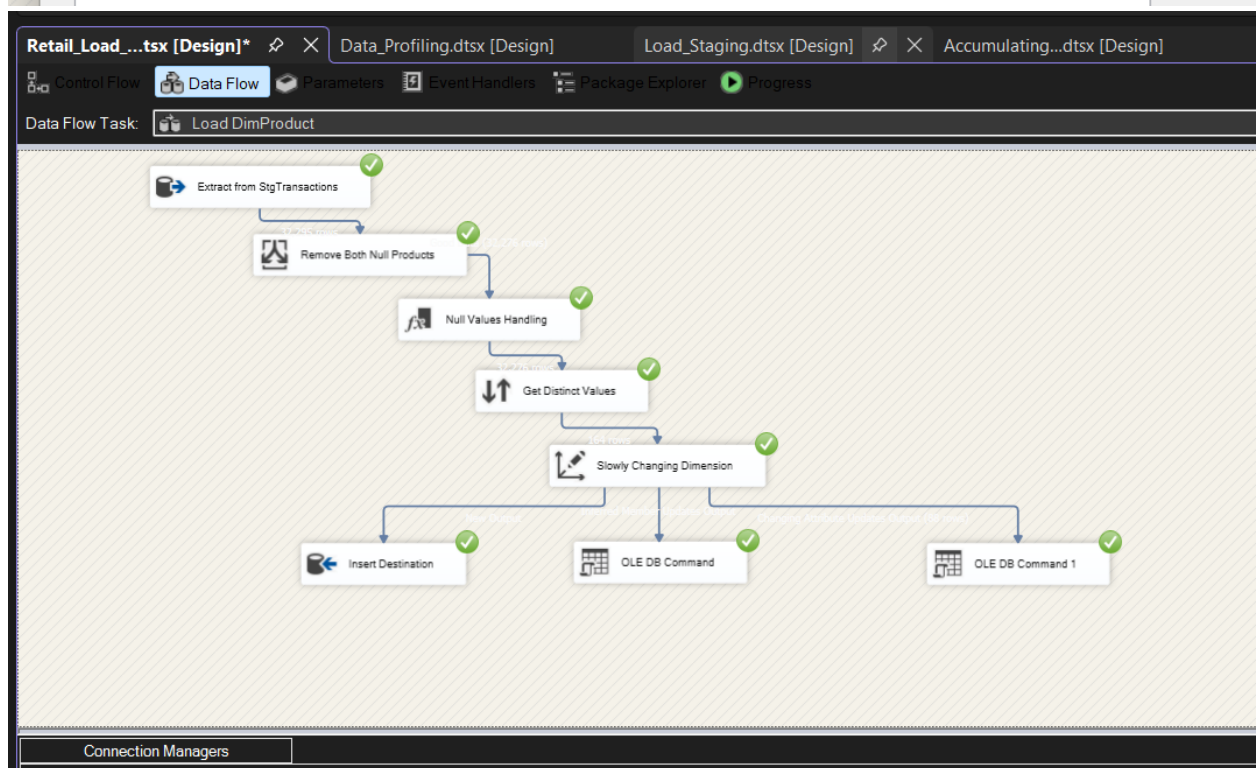
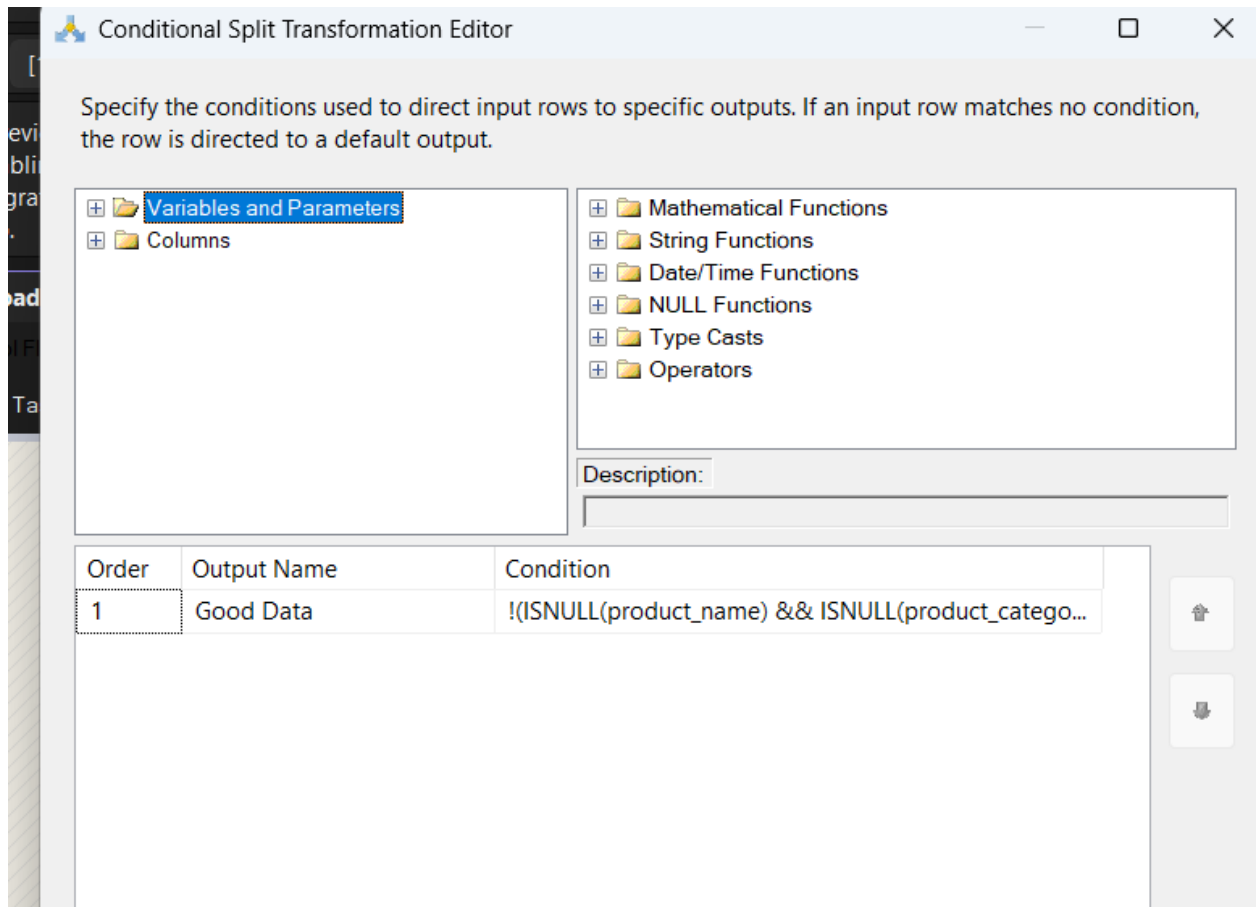




3.2.4 Conditional Splits

Remove null values using conditional split :

```
!(ISNULL(product_name) && ISNULL(product_category))
```



3.2.5 Sort Transformation

Use sort transformation to get distinct values

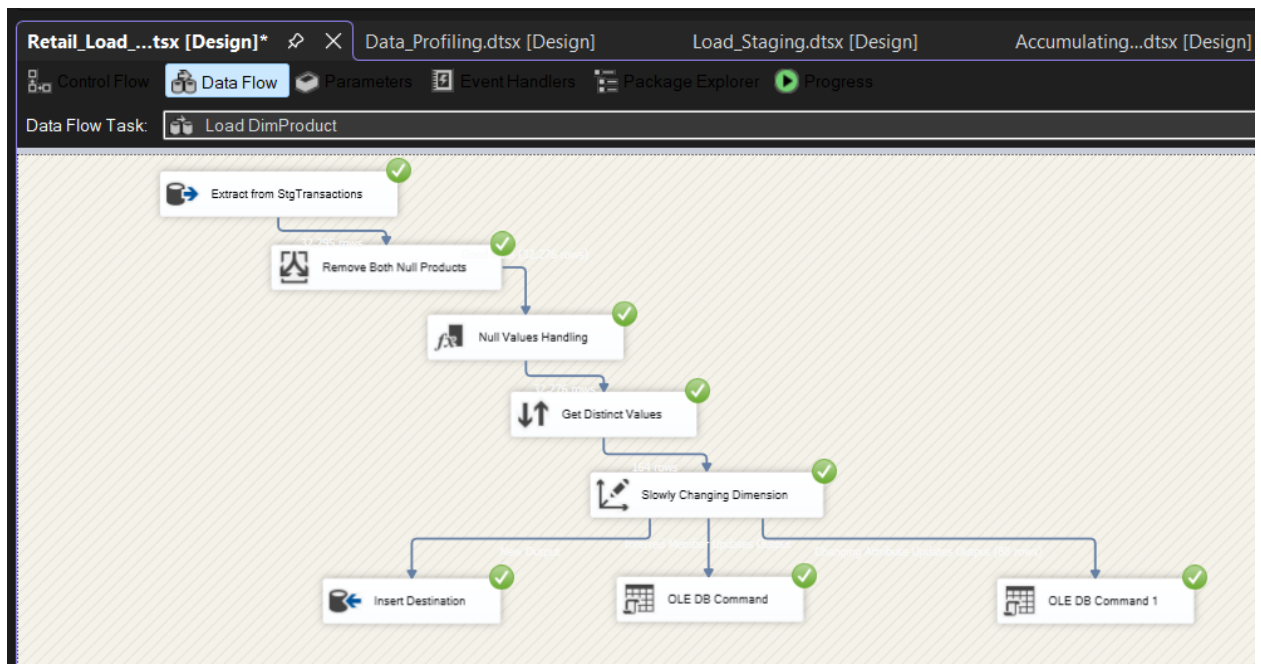
Sort Transformation Editor

Specify the columns to sort, and set their sort type and their sort order. All nonselected columns are copied unchanged.

Available Input Columns

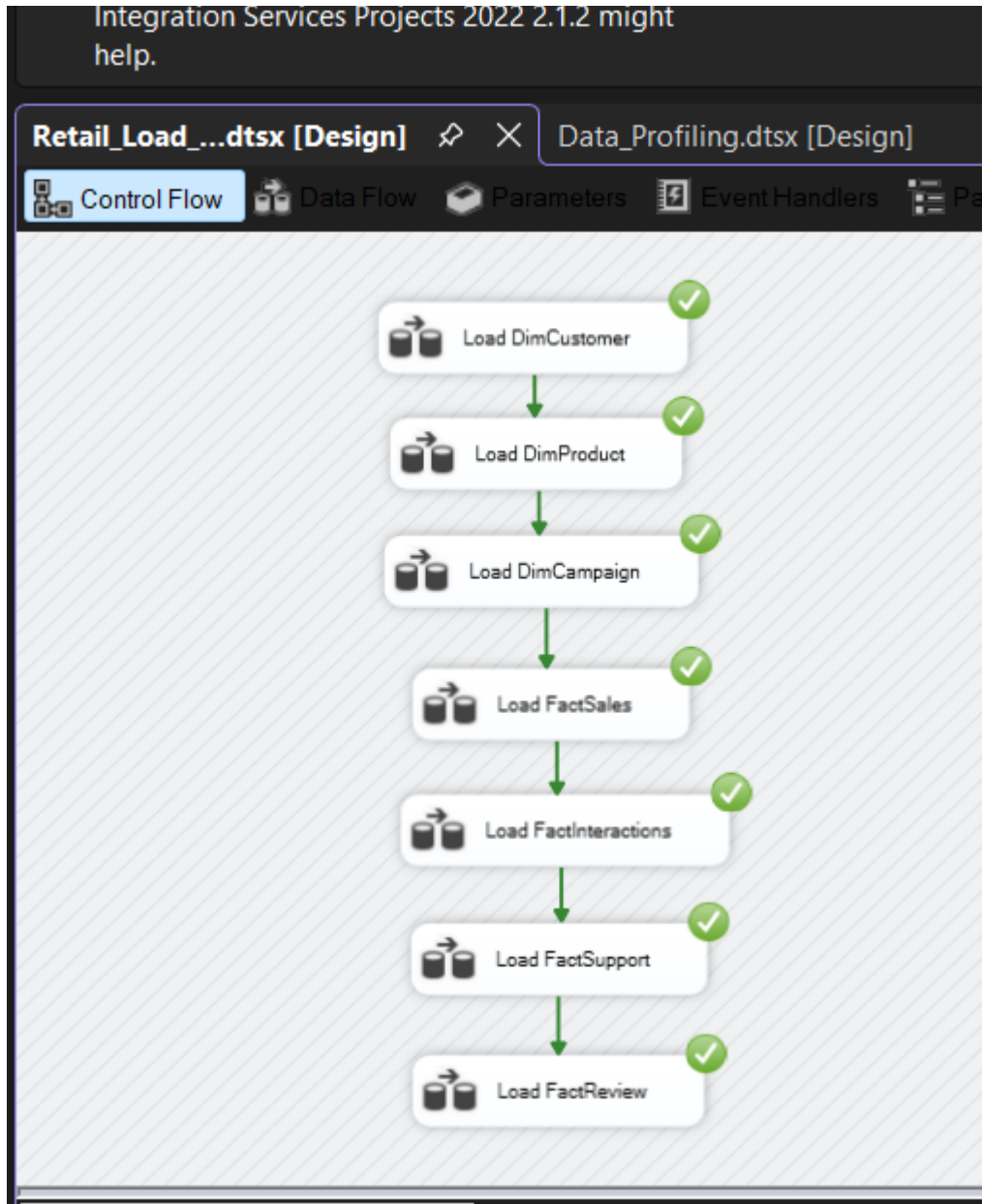
Available Input Columns	Pass Thr...
☒ Name	
☒ product_name	-

Input Column	Output Alias	Sort Type	Sort Order	Con
product_name	product_name	ascending	1	
product_category	product_category	ascending	2	

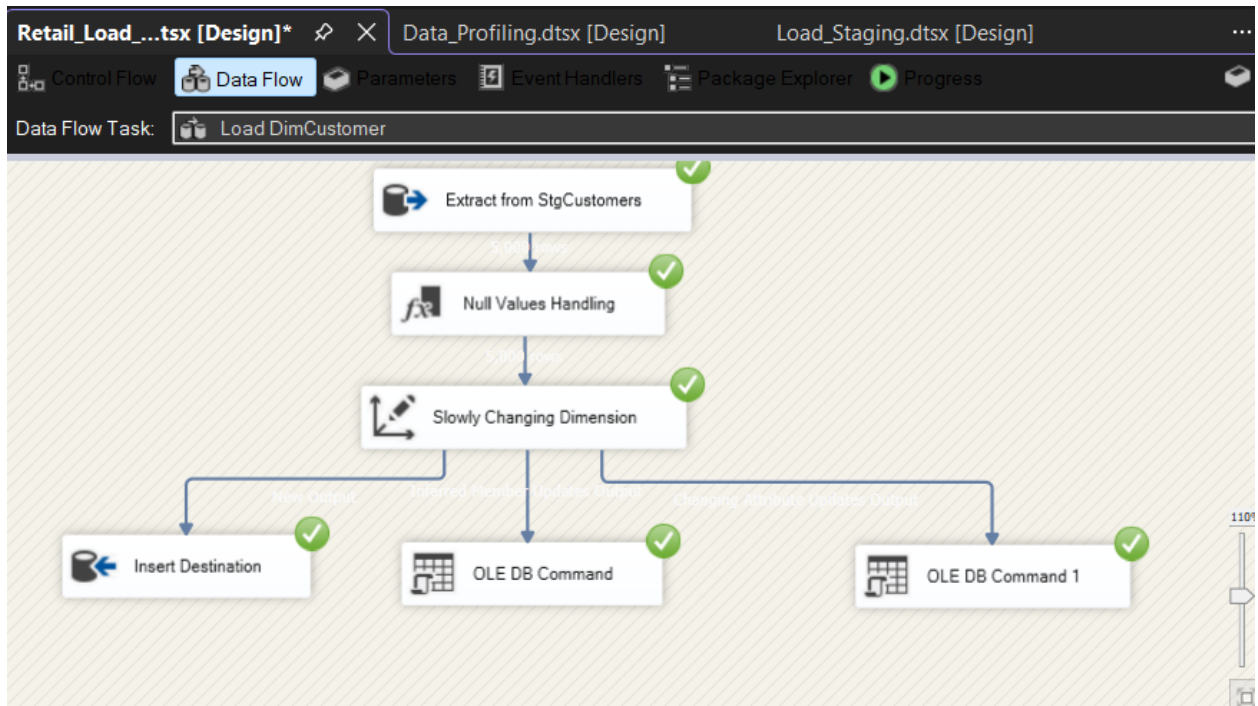


3.3 Data Loading (L)

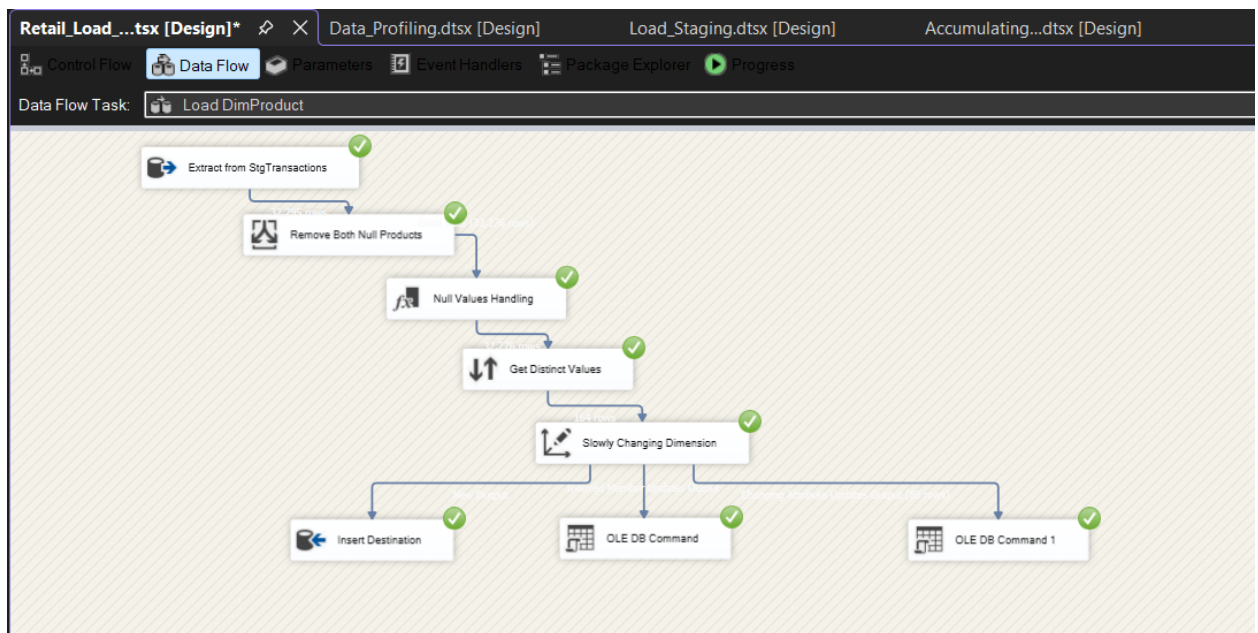
The final transformed data was loaded into the SQL Server Destination using the OLE DB Destination component.



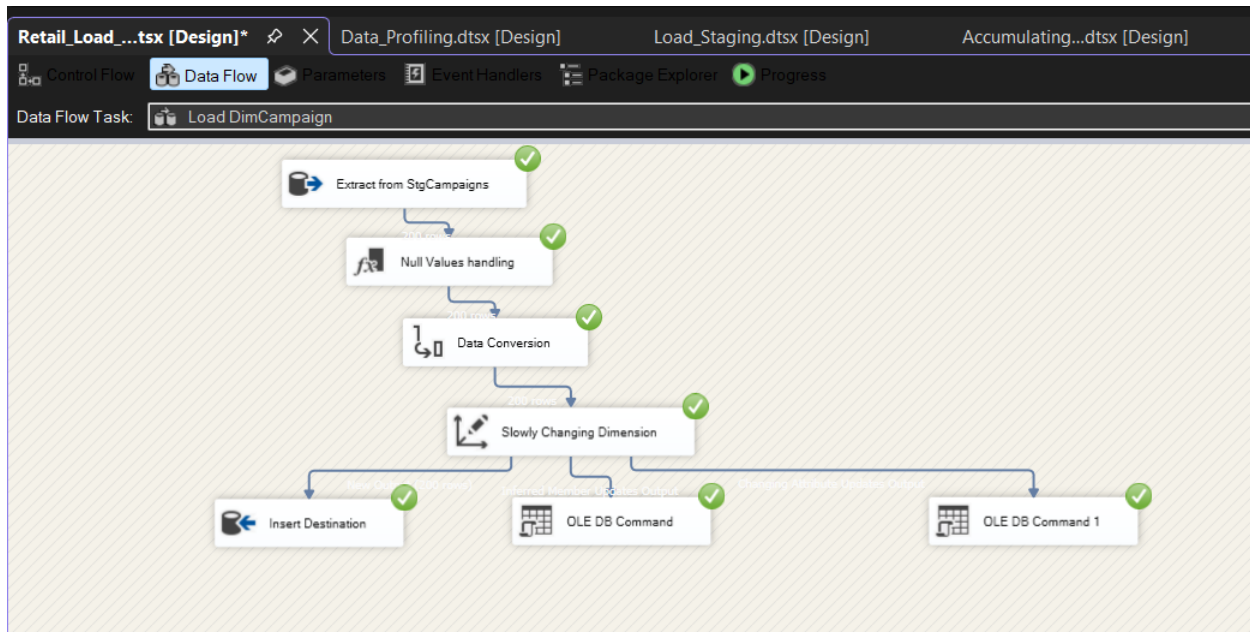
3.3.1.1 Load data to DimCustomer



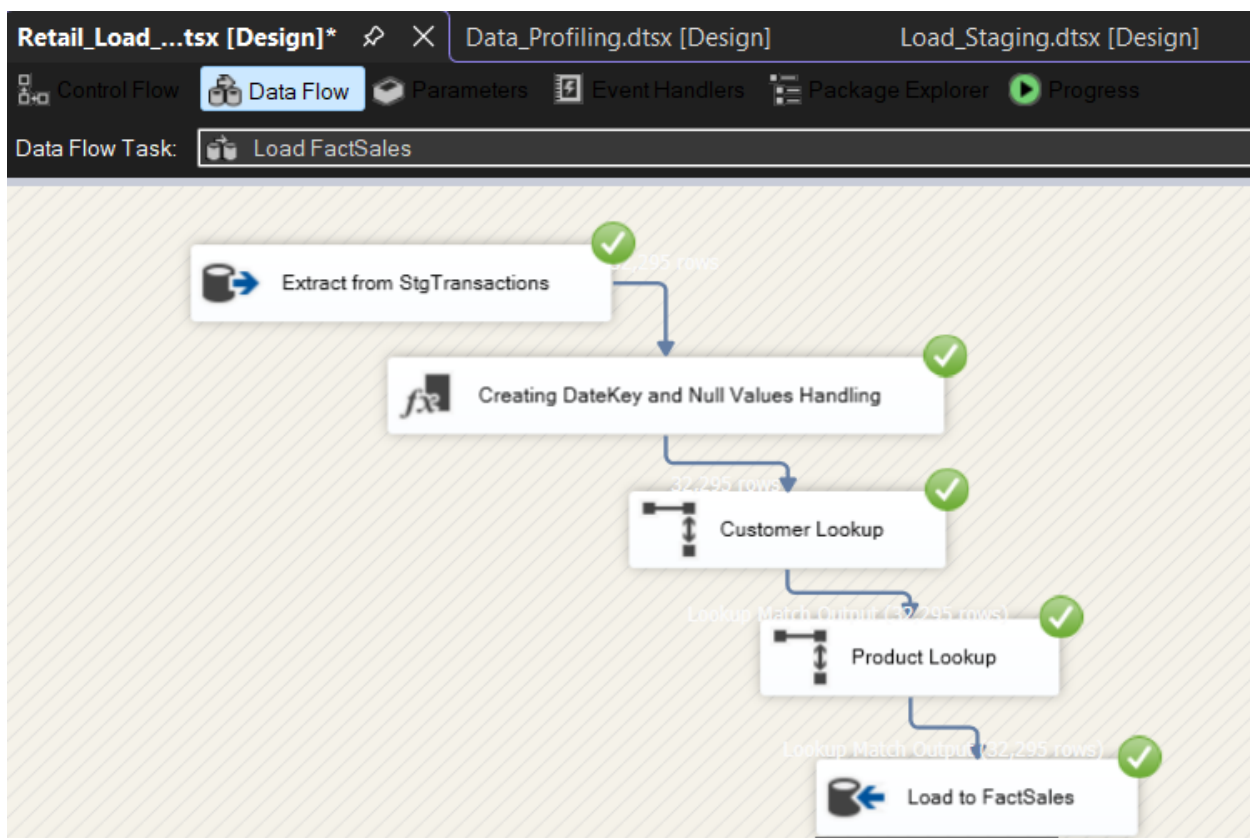
3.3.1.2 Load data to DimProduct



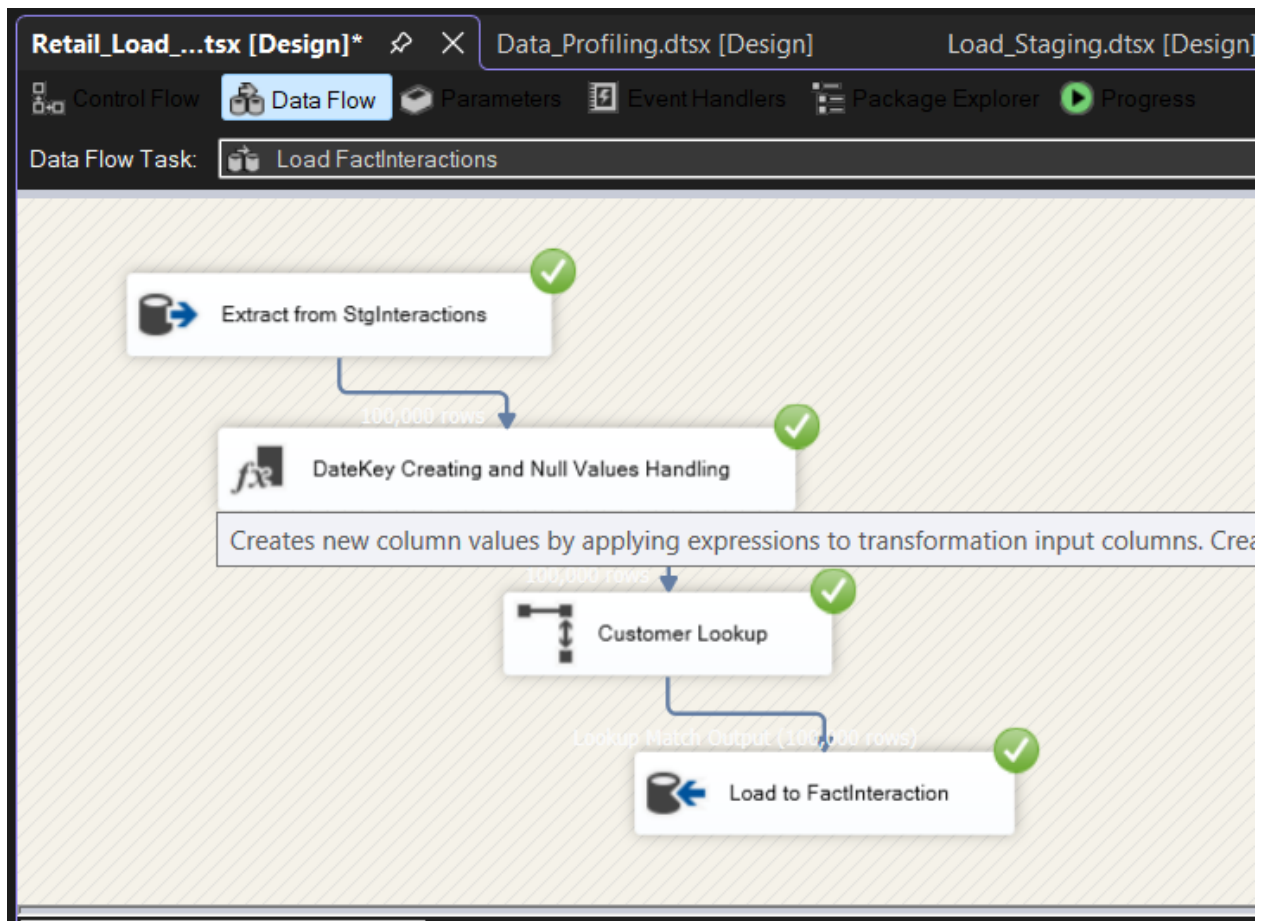
3.3.1.3 Load data to DimCampaign



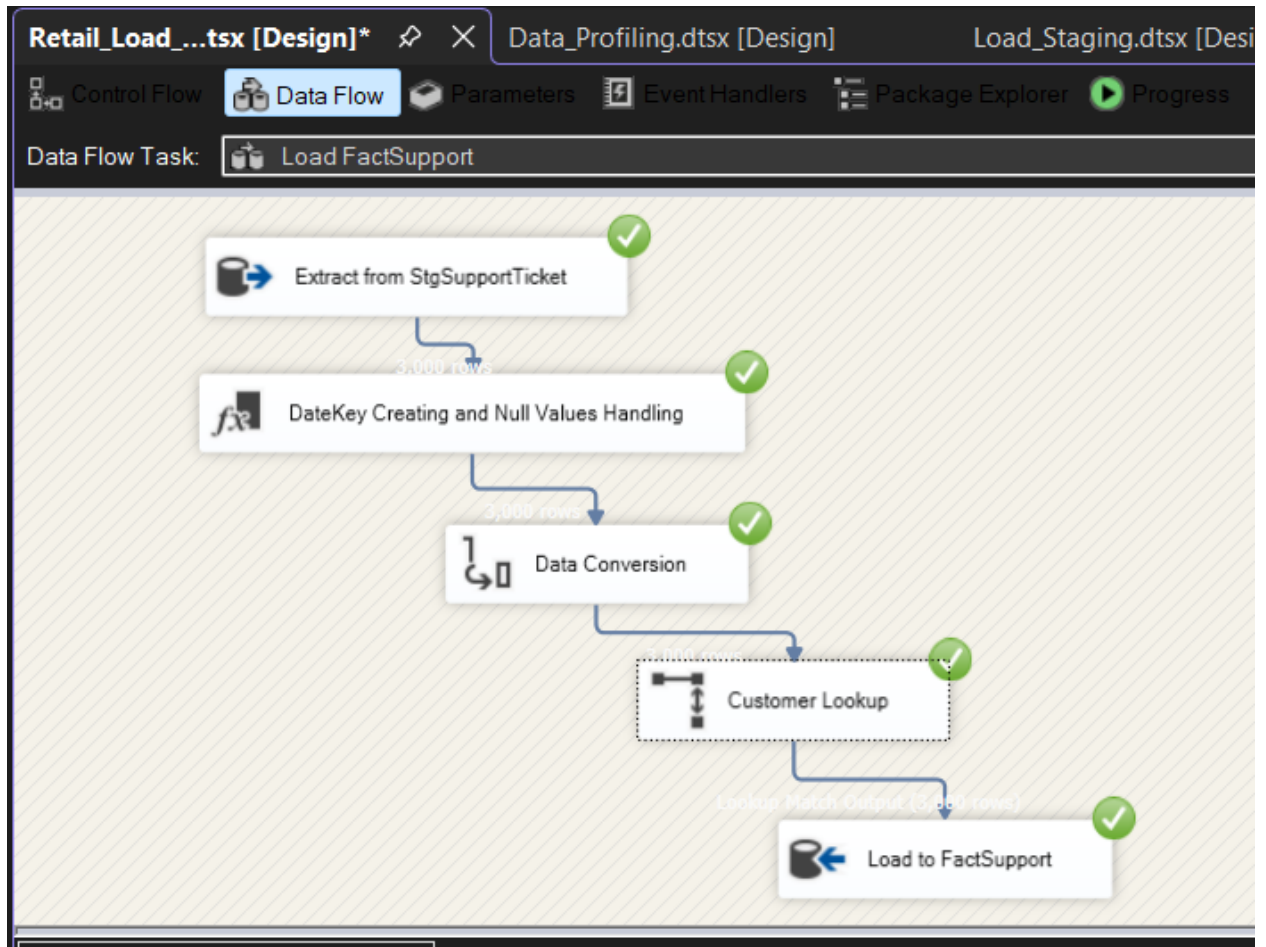
3.3.1.4 Load data to FactSales



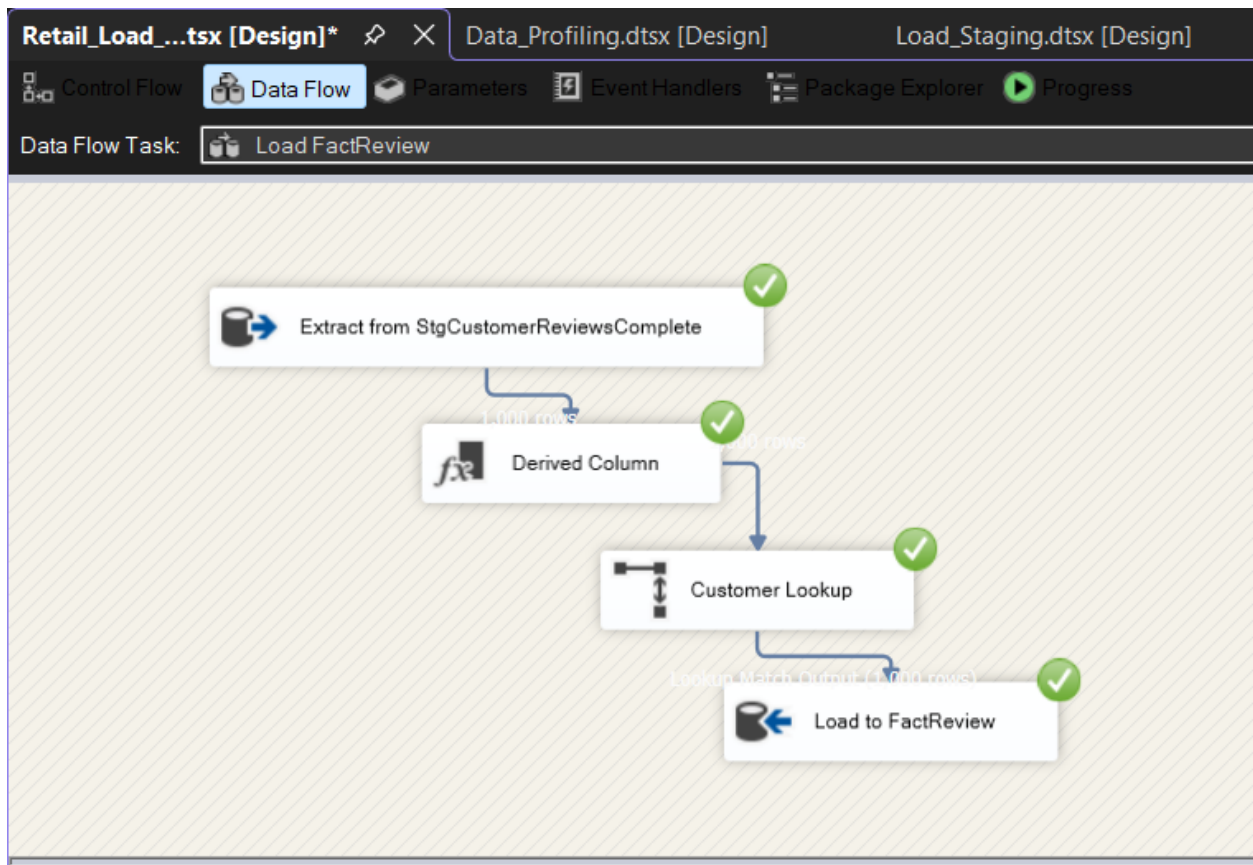
3.3.1.5 Load data to FactInteractions



3.3.1.6 Load data to FactSupport



3.3.1.7 Load data to FactReview



4 Accumulating Fact Update and Validation

4.1 Accumulating Fact Update Logic

4.1.1 Updating FactSales Table

The goal is to populate the `accm_txn_create_time`, `accm_txn_complete_time` and `txn_process_time_hours` columns in the FactSales table. This is based on the SalesDate and StoreLocation values.

4.1.1.1 Column: *accm_txn_create_time*

For all records the transaction creation time is set equal to the SalesDate. This is cast as a DATETIME value.

4.1.1.2 Column: *accm_txn_complete_time*

The transaction completion time is determined based on the StoreLocation:

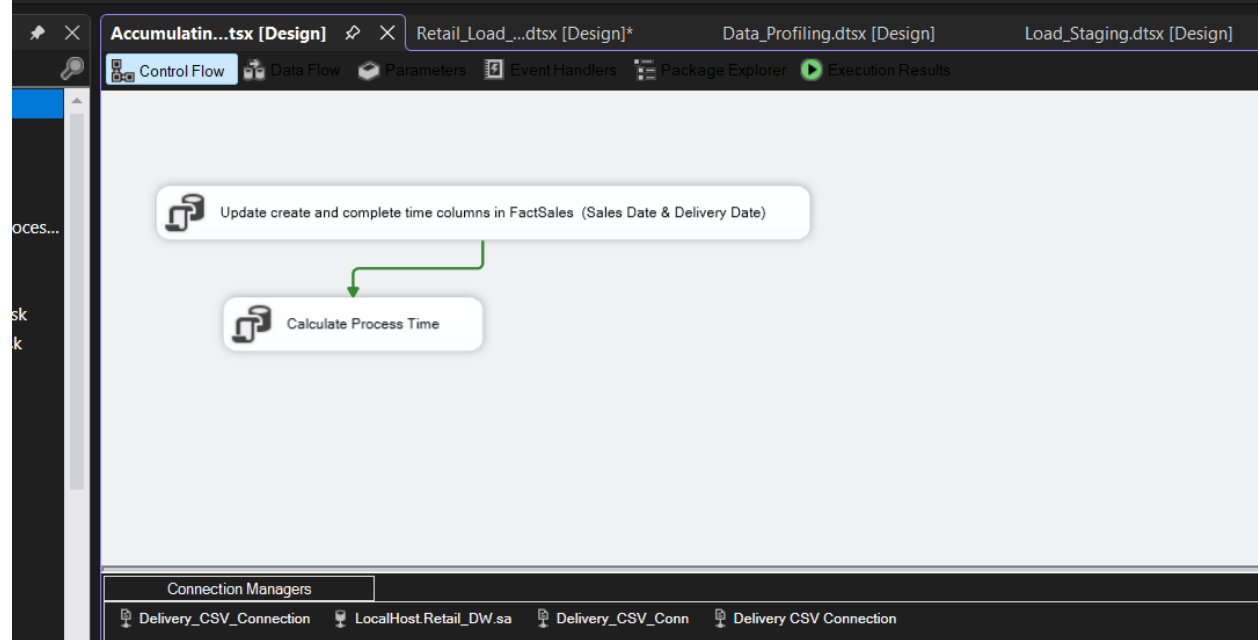
- If StoreLocation = 'Online'. The completion time is calculated by adding a number of days between 1 and 10 to the SalesDate.
- If StoreLocation is any value (e.g. In -Store). The completion time is set equal to the SalesDate.

4.1.1.3 Column: *txn_process_time_hours*

The processing time in hours is calculated as the difference between `accm_txn_complete_time` and `accm_txn_create_time`.

4.1.1.4 This logic was implemented in SSIS using two sequential Execute SQL Tasks.

ded unexpectedly. Disabling the extension SQL Server Integration Services Projects 2022 2.1.2 might help. [Disable this extension](#) [Manage performance](#) [Don't r](#)



4.1.1.5 Accumulated Fact Table

ToolsExtensionsWindowHelpSearch

SQLQuery10...W (sa (90))*SQLQuery6sq...DW (sa (86))*SQLQuery9sq...DW (sa (87))*DateMaster...DW (sa (79))SQLQuery5sq...DW (sa (81))*

1select * from FactSales;

100%No issues foundLn: 1, Ch: 1 (24 chars, 1 lines)TABS CRLF UTF-8

ResultsMessages

	CustomerKey	ProductKey	DateKey	Quantity	UnitPrice	DiscountApplied	TotalAmount	SalesDate	StoreLocation	PaymentMethod	accm_bn_create_time	accm_bn_complete_time	bn_process_time_hours	
1	z2ee	363	123	20201206	1	140.070007324219	20	120.07	2020-12-06	Houston, TX	Credit Card	2020-12-06 00:00:00.000	2020-12-06 00:00:00.000	0
2	le0c	947	107	20210426	1	339.570007324219	0	339.57	2021-04-26	Online	Gift Card	2021-04-26 00:00:00.000	2021-04-27 00:00:00.000	24
3	57d6d	1534	125	20211028	2	1335.88000488281	30	2641.76	2021-10-28	Online	PayPal	2021-10-28 00:00:00.000	2021-10-30 00:00:00.000	48
4	d9b8b	1646	49	20230316	1	91.5599975585938	25	66.56	2023-03-16	Online	Credit Card	2023-03-16 00:00:00.000	2023-03-26 00:00:00.000	240
5	0dab	3859	80	20220830	1	985.150024414063	0	985.15	2022-08-30	Atlanta, GA	Credit Card	2022-08-30 00:00:00.000	2022-08-30 00:00:00.000	0
6	b7ca3	4026	119	20211209	1	215.940002441406	0	215.94	2021-12-09	Boston, MA	Credit Card	2021-12-09 00:00:00.000	2021-12-09 00:00:00.000	0
7	ae8b	1594	119	20221224	1	989.549987792969	0	989.55	2022-12-24	Denver, CO	Gift Card	2022-12-24 00:00:00.000	2022-12-24 00:00:00.000	0
8	0958f	4973	66	20221030	1	227.360000610352	0	227.36	2022-10-30	Online	Debit Card	2022-10-30 00:00:00.000	2022-11-09 00:00:00.000	240
9	9645	1457	135	20250221	1	362.670013427734	0	362.67	2025-02-21	Miami, FL	Apple Pay	2025-02-21 00:00:00.000	2025-02-21 00:00:00.000	0
10	ae9f	2888	90	20240108	1	1033.26110839844	10	1023.26	2024-01-08	Online	Debit Card	2024-01-08 00:00:00.000	2024-01-10 00:00:00.000	48
11	tdfd	552	1	20220801	1	81.9599998044727	0	81.96	2022-08-01	Online	Debit Card	2022-08-01 00:00:00.000	2022-08-05 00:00:00.000	96
12	4ed3e	2984	7	20230611	1	230.940002441406	0	230.94	2023-06-11	Chicago, IL	Debit Card	2023-06-11 00:00:00.000	2023-06-11 00:00:00.000	0
13	308c	2507	19	20230420	3	22.7700004577637	0	68.31	2023-04-20	Online	Cash	2023-04-20 00:00:00.000	2023-04-29 00:00:00.000	216
14	f1d64	2079	157	20250224	1	188.429992675781	30	158.43	2025-02-24	Online	Credit Card	2025-02-24 00:00:00.000	2025-03-03 00:00:00.000	168
15	ae6d4	769	90	20221117	2	312.730010986328	15	610.46	2022-11-17	Online	Credit Card	2022-11-17 00:00:00.000	2022-11-20 00:00:00.000	72
16	8d8f0	3258	53	20241212	1	287	0	287.00	2024-12-12	Online	Gift Card	2024-12-12 00:00:00.000	2024-12-17 00:00:00.000	120
17	ae2b6	4139	33	20240723	1	235.080001031055	0	235.08	2024-07-23	Online	PayPal	2024-07-23 00:00:00.000	2024-07-27 00:00:00.000	96
18	ae843	2232	13	20230422	1	63.5699996048242	0	63.57	2023-04-22	Online	Debit Card	2023-04-22 00:00:00.000	2023-05-01 00:00:00.000	216
19	43b16	3904	105	20240619	2	196.029998779297	0	392.06	2024-06-19	Online	PayPal	2024-06-19 00:00:00.000	2024-06-26 00:00:00.000	168
20	986e	2412	145	20240618	2	173.690002441406	0	347.38	2024-06-18	Online	PayPal	2024-06-18 00:00:00.000	2024-06-25 00:00:00.000	168
21	icoc	1544	113	20241108	1	222.5	0	222.50	2024-11-08	Online	Debit Card	2024-11-08 00:00:00.000	2024-11-13 00:00:00.000	120
22	577dd	4233	163	20230511	2	564.780029296875	25	1104.56	2023-05-11	Online	N/A	2023-05-11 00:00:00.000	2023-05-20 00:00:00.000	216
23	2e42	950	7	20231213	1	130.759994506836	0	130.76	2023-12-13	Atlanta, GA	PayPal	2023-12-13 00:00:00.000	2023-12-13 00:00:00.000	0
24	9421	1304	80	20221219	1	1640.56005859375	20	1620.56	2022-12-19	Online	Debit Card	2022-12-19 00:00:00.000	2022-12-27 00:00:00.000	192
25	118cb	310	139	20210403	1	2634.27001953125	0	2634.27	2021-04-03	Online	Credit Card	2021-04-03 00:00:00.000	2021-04-05 00:00:00.000	48
26	51d6	1357	161	20210806	1	0	0	0.00	2021-08-06	Online	PayPal	2021-08-06 00:00:00.000	2021-08-11 00:00:00.000	120
27	438ea	2646	157	20230207	1	293.109885351563	0	293.11	2023-02-07	Miami, FL	Cash	2023-02-07 00:00:00.000	2023-02-07 00:00:00.000	0
28	88aa8	3574	111	20241224	3	649.809885351563	20	1928.83	2024-12-24	Chicago, IL	Gift Card	2024-12-24 00:00:00.000	2024-12-24 00:00:00.000	0
29	cae9	1699	149	20250106	1	273.839996337891	0	273.84	2025-01-06	Seattle, WA	Credit Card	2025-01-06 00:00:00.000	2025-01-06 00:00:00.000	0
30	37385	2091	133	20240723	1	112.610000610352	25	87.61	2024-07-23	Atlanta, GA	Apple Pay	2024-07-23 00:00:00.000	2024-07-23 00:00:00.000	0
31	icd470	4626	155	20210707	1	2950.46997070313	0	2950.47	2021-07-07	Online	Credit Card	2021-07-07 00:00:00.000	2021-07-11 00:00:00.000	96
32	350d	1031	21	20230513	1	419.290008544922	0	419.29	2023-05-13	Online	Credit Card	2023-05-13 00:00:00.000	2023-05-14 00:00:00.000	24
33	1a7d	4419	76	20221211	2	1220.26000026563	0	2440.52	2022-12-11	Online	Cash	2022-12-11 00:00:00.000	2022-12-15 00:00:00.000	96

SQLQuery10....W (sa (90)) * X SQLQuery6.sql...DW (sa (86)) * SQLQuery9.sql...DW (sa (87)) * DateMaster....DW (sa (79)) SQLQuery5.sql...DW (sa (81)) *

1 | select * from FactSales;

100 % No issues found Lr: 1, Ch: 1 (24 chars, 1 lines) TABS CRLF UTF-8

		CustomerKey	ProductKey	DateKey	Quantity	UnitPrice	DiscountApplied	TotalAmount	SalesDate	StoreLocation	PaymentMethod	accm_txn_create_time	accm_txn_complete_time	txn_process_time_hours
82	i926d8	2858	111	20250117	1	1216.35998535156	25	1191.36	2025-01-17	Online	Apple Pay	2025-01-17 00:00:00.000	2025-01-23 00:00:00.000	144
83	d2ef7	4606	123	20230526	2	147.240005493164	0	294.48	2023-05-26	Chicago, IL	Debit Card	2023-05-26 00:00:00.000	2023-05-26 00:00:00.000	0
84	i5f9d2	4241	161	20221024	1	271.890014648316	20	251.89	2022-10-24	Online	PayPal	2022-10-24 00:00:00.000	2022-11-02 00:00:00.000	216
85	3525	3906	39	20231024	1	266.679992675781	0	266.68	2023-10-24	Online	Debit Card	2023-10-24 00:00:00.000	2023-10-31 00:00:00.000	168
86	hf7	3814	47	20241221	1	463.269898013672	0	463.27	2024-12-21	Online	Credit Card	2024-12-21 00:00:00.000	2024-12-27 00:00:00.000	144
87	ib88d8	1143	17	20241019	1	1192.5600589375	0	1192.56	2024-10-19	Miami, FL	Credit Card	2024-10-19 00:00:00.000	2024-10-19 00:00:00.000	0
88	iax33	1135	41	20231104	1	1040.63000488281	0	1040.63	2023-11-04	Online	Google Pay	2023-11-04 00:00:00.000	2023-11-07 00:00:00.000	72
89	32e3e	2396	90	20230602	1	300.85998531563	5	295.86	2023-06-02	Seattle, WA	Credit Card	2023-06-02 00:00:00.000	2023-06-02 00:00:00.000	0
90	b4t	1055	33	20240918	2	119.11000610352	0	238.22	2024-09-18	Online	Credit Card	2024-09-18 00:00:00.000	2024-09-21 00:00:00.000	72
91	iecf8	1895	125	20210722	1	756.119995117188	30	726.12	2021-07-22	Online	PayPal	2021-07-22 00:00:00.000	2021-07-24 00:00:00.000	48
92	ibda	4586	1	20241122	1	140.759994508636	15	125.76	2024-11-22	Online	Credit Card	2024-11-22 00:00:00.000	2024-11-23 00:00:00.000	24
93	r304	3730	86	20230725	2	457.230010986328	0	914.46	2023-07-25	Online	Debit Card	2023-07-25 00:00:00.000	2023-08-04 00:00:00.000	240
94	iccd2	1849	27	20240830	1	135.759994508636	5	130.76	2024-08-30	Online	Credit Card	2024-08-30 00:00:00.000	2024-09-08 00:00:00.000	216
95	343c9	1613	151	20240815	1	183.550003051758	0	183.55	2024-08-15	San Francisco, CA	Credit Card	2024-08-15 00:00:00.000	2024-08-15 00:00:00.000	0
96	1c	4588	51	20230904	1	179.380004882813	0	179.38	2023-09-04	Online	Credit Card	2023-09-04 00:00:00.000	2023-09-09 00:00:00.000	120
97	8b205	4421	157	20240114	1	234.339996337891	0	234.34	2024-01-14	Chicago, IL	Debit Card	2024-01-14 00:00:00.000	2024-01-14 00:00:00.000	0
98	e6ae	3021	33	20220713	1	211.699996948242	10	201.70	2022-07-13	Denver, CO	PayPal	2022-07-13 00:00:00.000	2022-07-13 00:00:00.000	0
99	H090	3967	7	20230816	1	197.830001831055	0	197.83	2023-08-16	Online	Debit Card	2023-08-16 00:00:00.000	2023-08-22 00:00:00.000	144
100	bab0f	3324	163	20230331	1	1071.39001464844	0	1071.39	2023-03-31	Online	Debit Card	2023-03-31 00:00:00.000	2023-04-10 00:00:00.000	240
101	264d0	3701	107	20241209	2	315.899993896484	15	616.80	2024-12-09	Online	Debit Card	2024-12-09 00:00:00.000	2024-12-15 00:00:00.000	144
102	3f642	670	39	20231004	1	469.309997558594	0	469.31	2023-10-04	Los Angeles, CA	Credit Card	2023-10-0		

5 Data Validation

To make sure the ETL process worked well we ran a SQL query in SQL Server Management Studio.

5.1 Row count validating

Customers tables

The screenshot shows the SQL Server Management Studio interface. The query editor displays the following SQL query:

```
1 SELECT COUNT(*) AS SourceCount FROM Retail_SourceDB.dbo.Customers;  
2  
3 SELECT COUNT(*) AS StagingCount FROM Retail_Staging.dbo.StgCustomers;  
4  
5 SELECT COUNT(*) AS DimCount FROM Retail_DW.dbo.DimCustomer;  
6
```

The Results pane shows the output of the query, which consists of three rows of counts:

SourceCount
1
5000

StagingCount
1
5000

DimCount
1
5000

5.1.1.1 FactSales Table

The screenshot shows the SQL Server Management Studio interface. The query editor displays the following SQL query:

```
1 SELECT COUNT(*) AS SourceCount FROM Retail_Staging.dbo.StgTransactions;  
2  
3 SELECT COUNT(*) AS TargetCount FROM Retail_DW.dbo.FactSales;
```

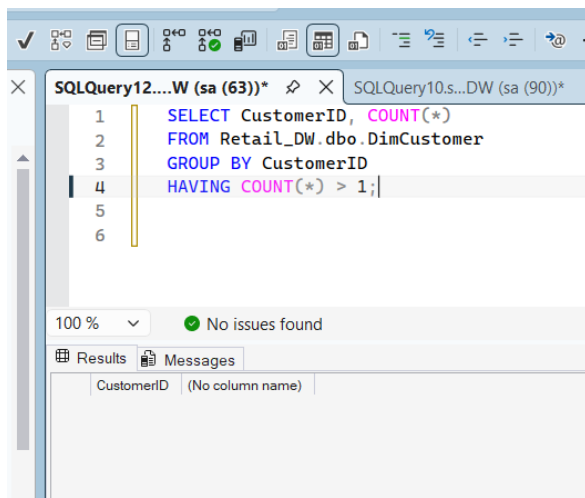
The Results pane shows the output of the query, which consists of two rows of counts:

SourceCount
1
32295

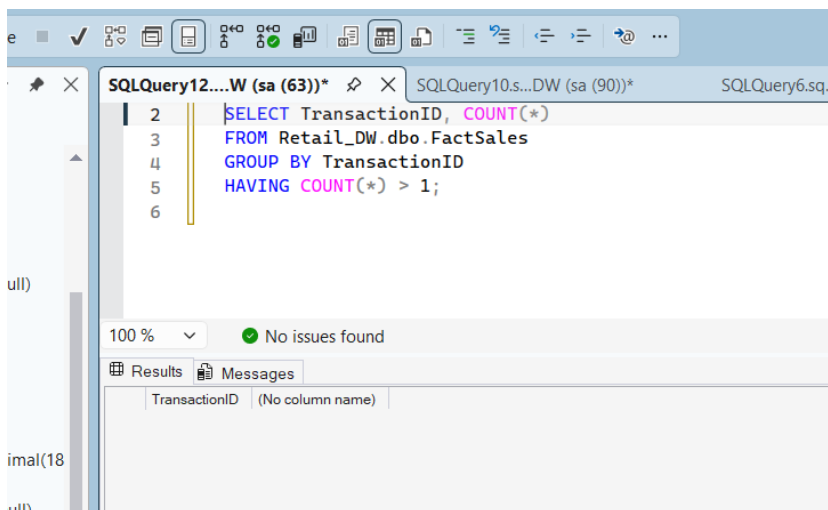
TargetCount
1
32295

5.2 Duplicate Check

5.2.1.1 Customers table duplicate columns checking



5.2.1.2 FactSales table duplicate columns checking



6 Conclusion

This project showed how to build a Data Warehouse from start to finish.

I took a retail dataset designed a simple Star Schema and used SSIS to move the data.

This made the operational data useful for business decisions.

The ETL process and Data Warehouse design helped turn data into a resource.